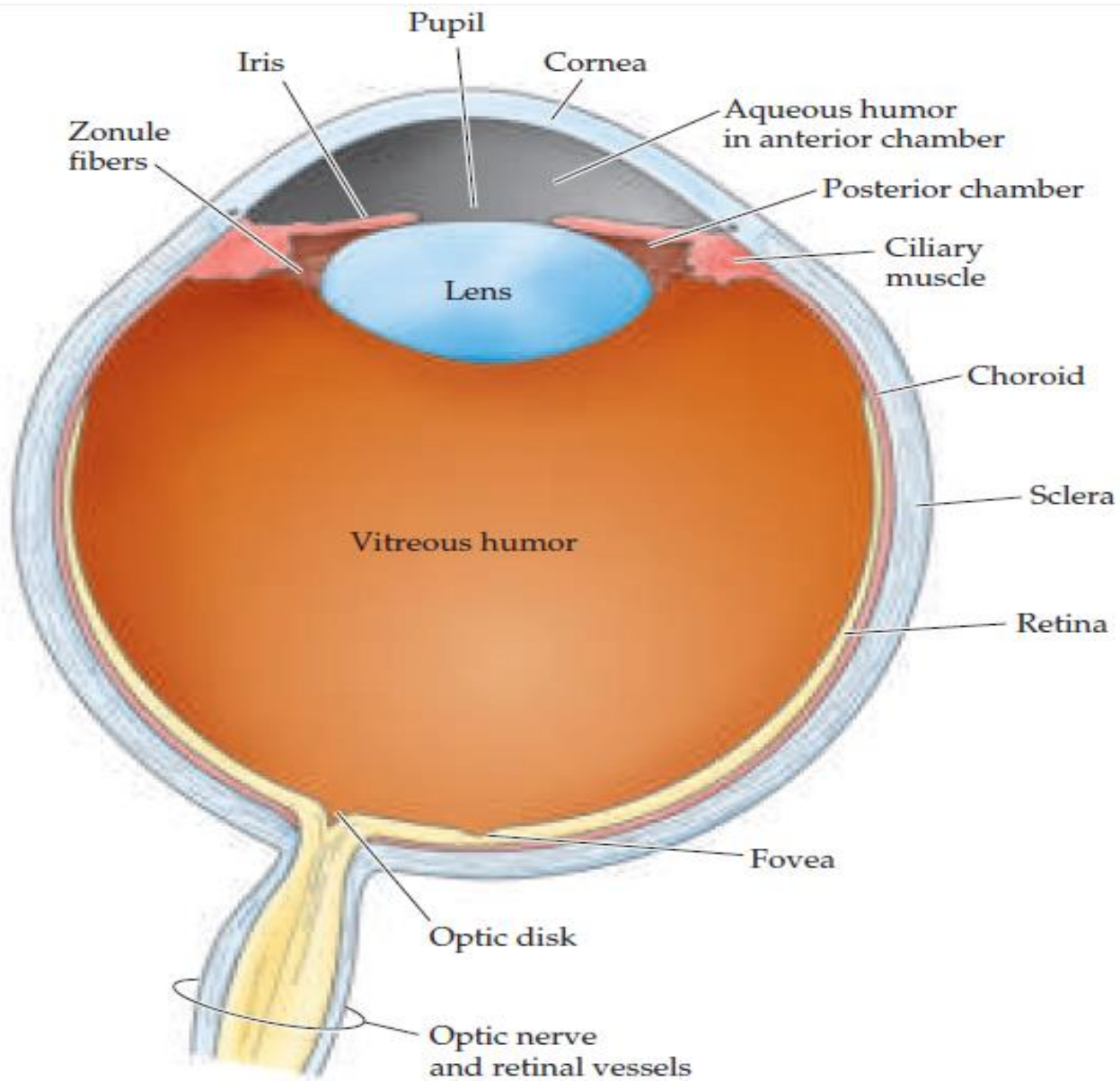

The Physiology of Vision

The Eye

- Structural components:
 1. Sclera (*white of the eye*)
 2. Cornea
 3. Choroid
 4. Ciliary body
 5. Iris + Pupil
 6. Retina
 7. Lens
 8. Fluid environments → Aqueous & Vitreous humor

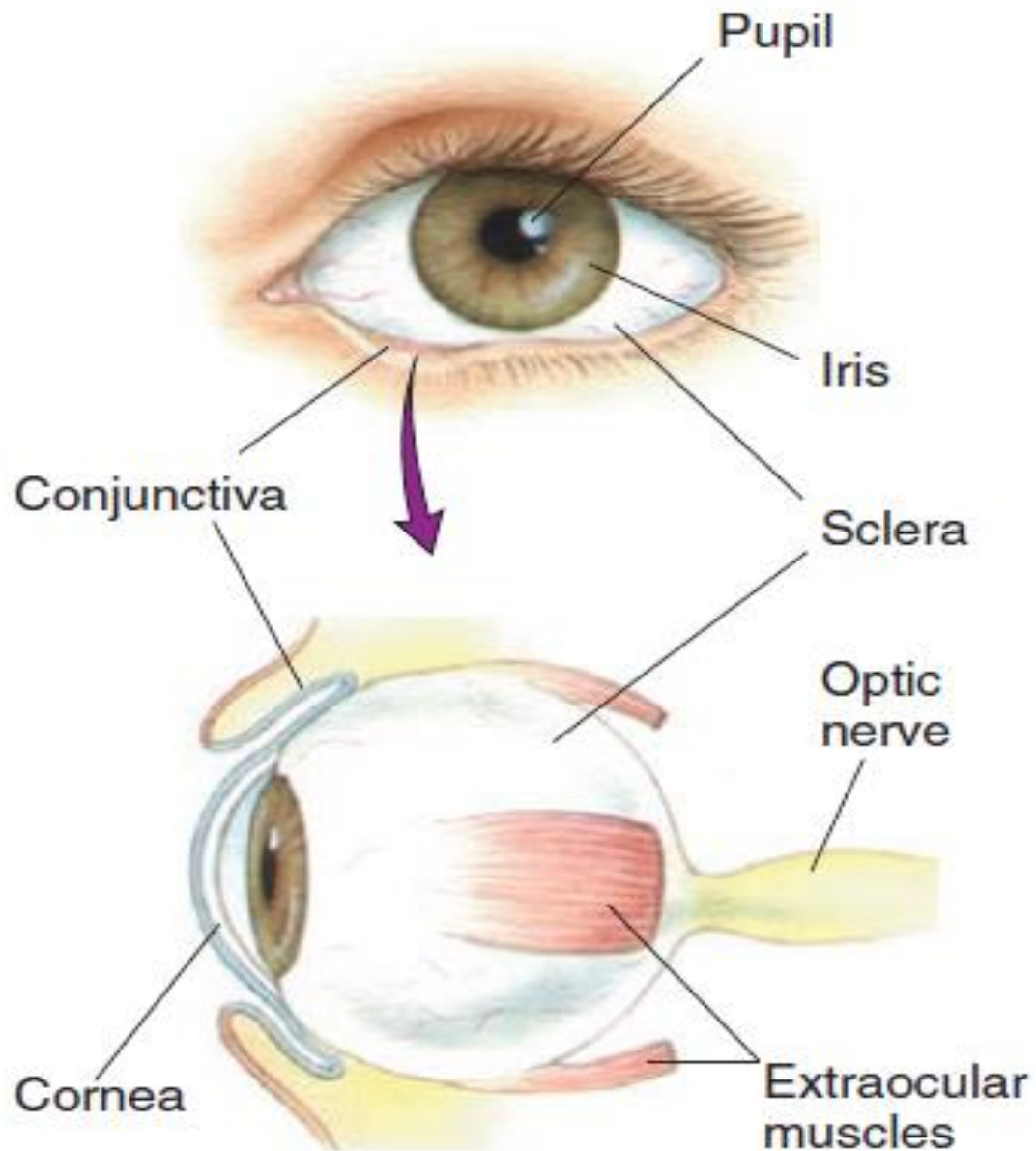


The Eye

- Sclera
 - A strong outer layer of dense white fibrous tissue (*opaque layer*)
 - *Protects the inner layers*
 - Extraocular muscles (3 pairs) are inserted into the sclera which position the eye (*move the eyeball in the orbit*)
 - Sclera is transformed into the cornea (*specialized transparent tissue*) at the front of the eye

The Eye

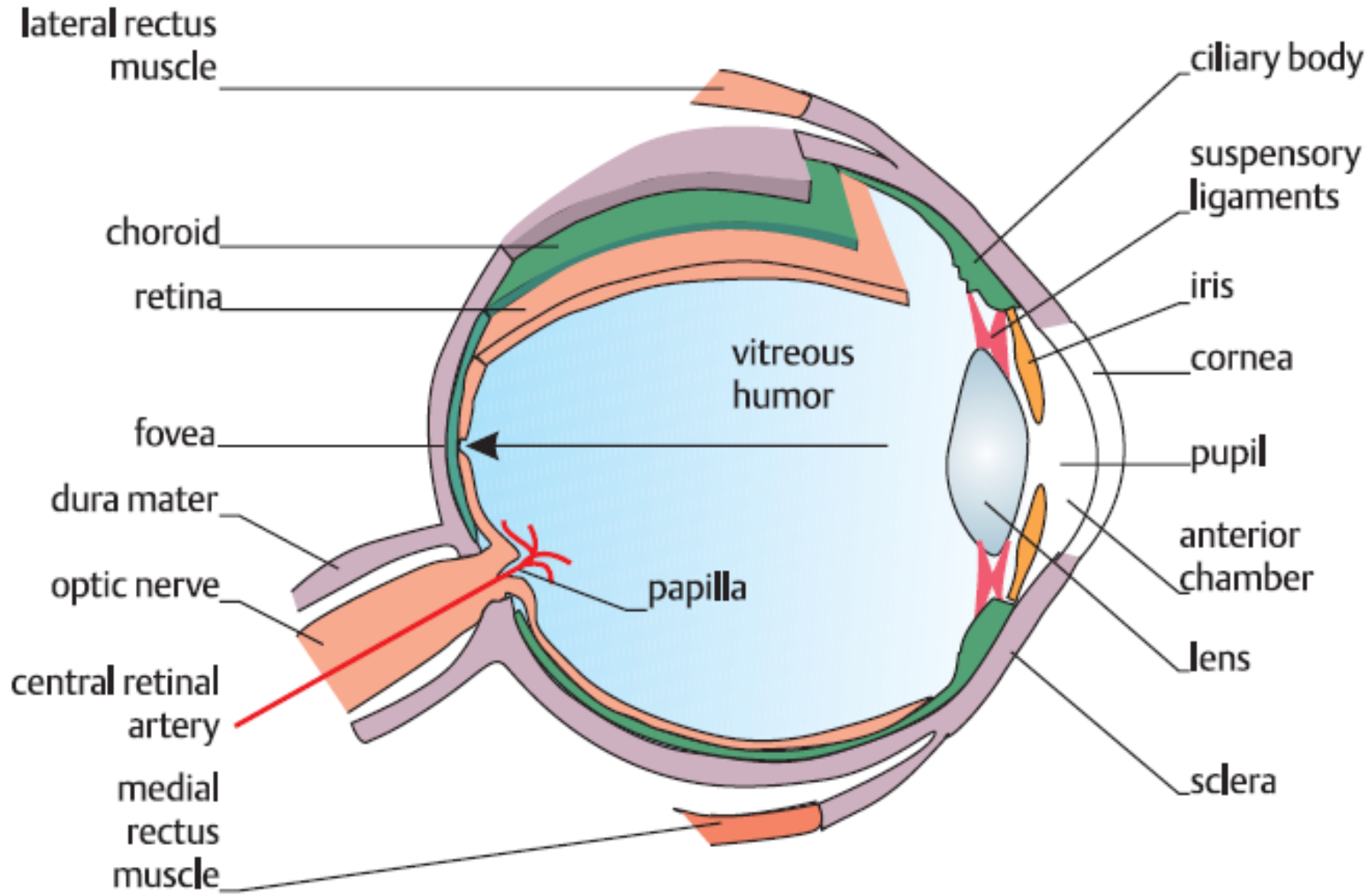
- Cornea:
 - Anterior non-vascularized, thin transparent portion of the sclera
 - Nourished by the fluid behind it (*the aqueous humor*)
 - Allow light rays to enter the eye
 - Highly innervated with unmyelinated fibers → sensitive to touch
 - Lateral border of the cornea is continuous with the conjunctiva
- Conjunctiva:
 - A membrane that folds back from the inside of the eyelids and attaches to the sclera



The Eye

- Choroid
 - Located inside the sclera
 - Composed of a **rich capillary bed** → nourish structures of the eye
 - High concentration of **melanin** (*light absorbing pigment*) → prevent penetration of light rays
 - Thicken and form rings (*ciliary body*) at the front of the eye

Cross section of the Eye



The Eye

- Ciliary body:
 - A ring of tissue that encircles the lens
 - Ciliary processes (*vascular component*) → produces fluid (*aqueous humor*) that fills the posterior & anterior chamber
 - Ciliary muscles → adjust refractive power of the lens by controlling its curvature (*accommodation*)
 - Ciliary muscles are attached to zonule fibers (*radially arranged connective tissue bands*) which hold the lens in place

The Eye

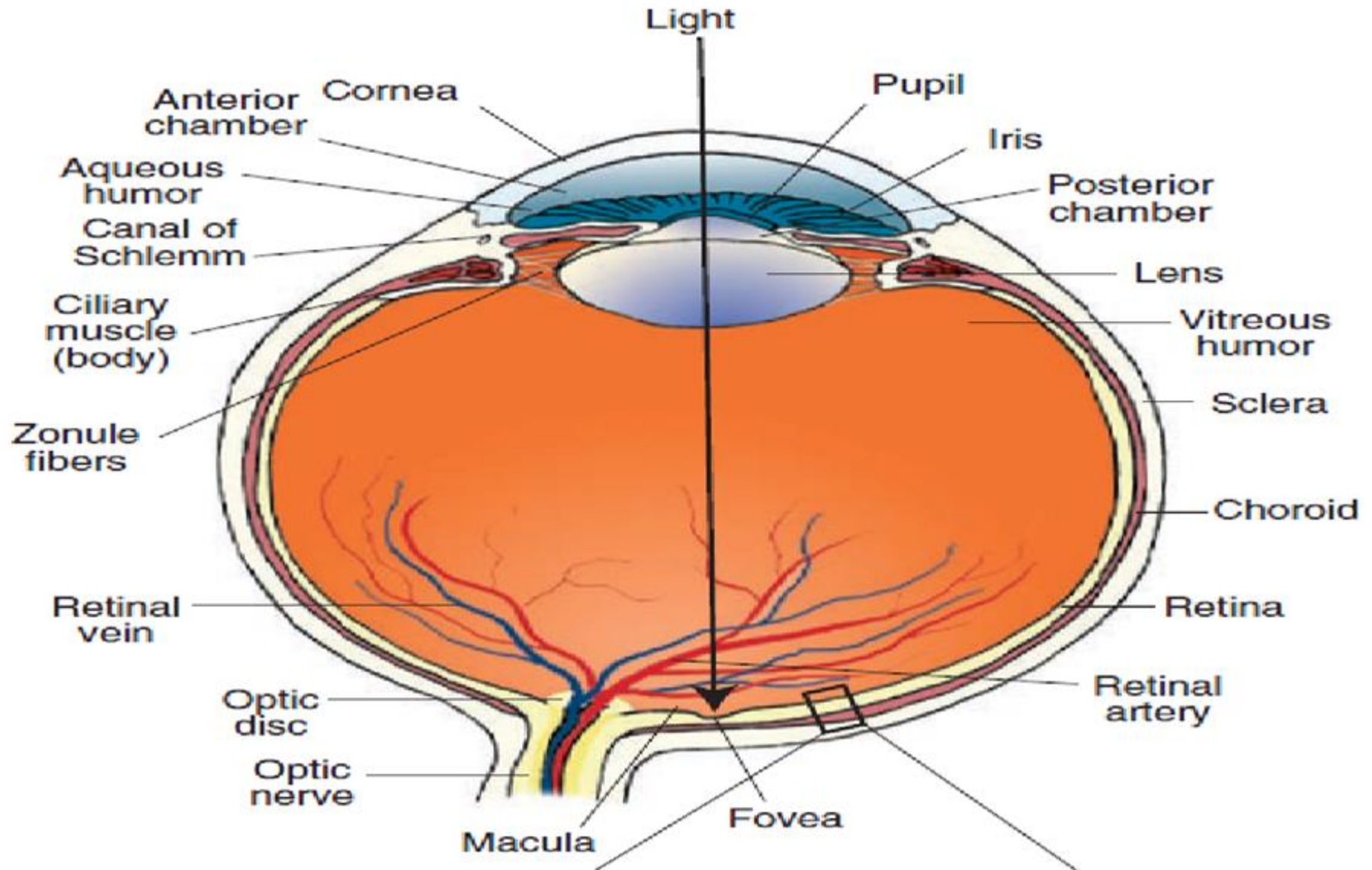
- Lens:
 - A transparent crystalline structure held in place by suspensory ligaments (zonular fibers) situated just behind the pupil
 - Allow light rays to pass through
 - Refract and focus light rays on to the retina
 - Accommodation → the change in refractive power of the lens

The Eye

- Iris:
 - Colored portion of the eye, visible through the cornea
 - Pupil → central opening of iris
 - Contain circular & radial smooth muscles with opposing actions
→ adjust the size of the pupil under neural control
 - PNS stimulation → contraction of circular muscles → constriction of the pupil → for near vision
 - SNS stimulation → contraction of radial muscles → dilation of the pupil → for far vision

The Eye

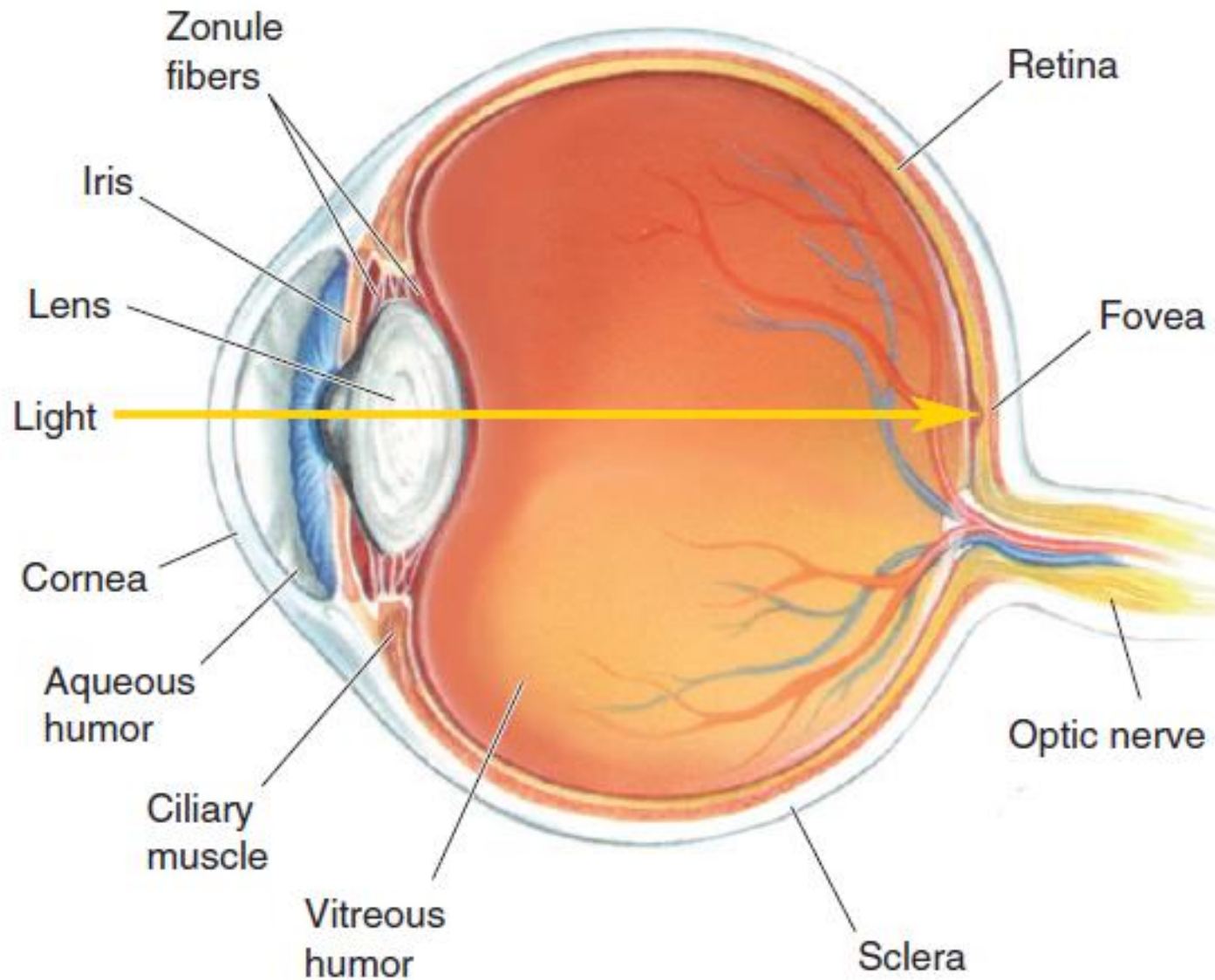
- Retina
 - Innermost layer which contain **photoreceptors** that are sensitive to light and are capable of transducing visual signals
- Optic disc → entry point for optic nerve + retinal artery
 - *The blind spot* → no photoreceptors in the optic disc
- Macula:
 - **Yellow spot** at the middle of each retina lateral to the optic disc
 - Relative absence of large blood vessels
 - For **central vision** (*as opposed to peripheral vision*)



Cross section of the Eye

The Eye

- Fovea:
 - A shallow depression at the center of the macula, devoid of blood vessels
 - Contain only cones
 - Region of the retina with minimum scattering of light rays (highest visual acuity) due to:
 - Absence of layers of cell bodies and their processes +
 - Diversion of blood vessels away from the fovea



Cross section of the Eye

The Eye

- Aqueous Humor
 - Clear, watery fluid produced continuously by the **ciliary processes** in the **posterior chamber** (*space between the lens and the iris*)
 - Flows through the pupil into the **anterior chamber** (*space in front of the lens, behind the cornea*)
 - Similar in composition with plasma, but with $\approx 1\%$ of its protein
 - Provide nutrition to the lens and cornea
 - Reabsorbed through **trabeculae** (*specialized collection of cells*) into the **canal of Schlemm** (*venous channel*) at the junction of iris & cornea

The Eye

- Aqueous Humor
 - Inadequate drainage of aqueous humor → excessive accumulation of fluid → ↑IOP → reduces blood supply to the eye → damage to the retina → ***Glaucoma***

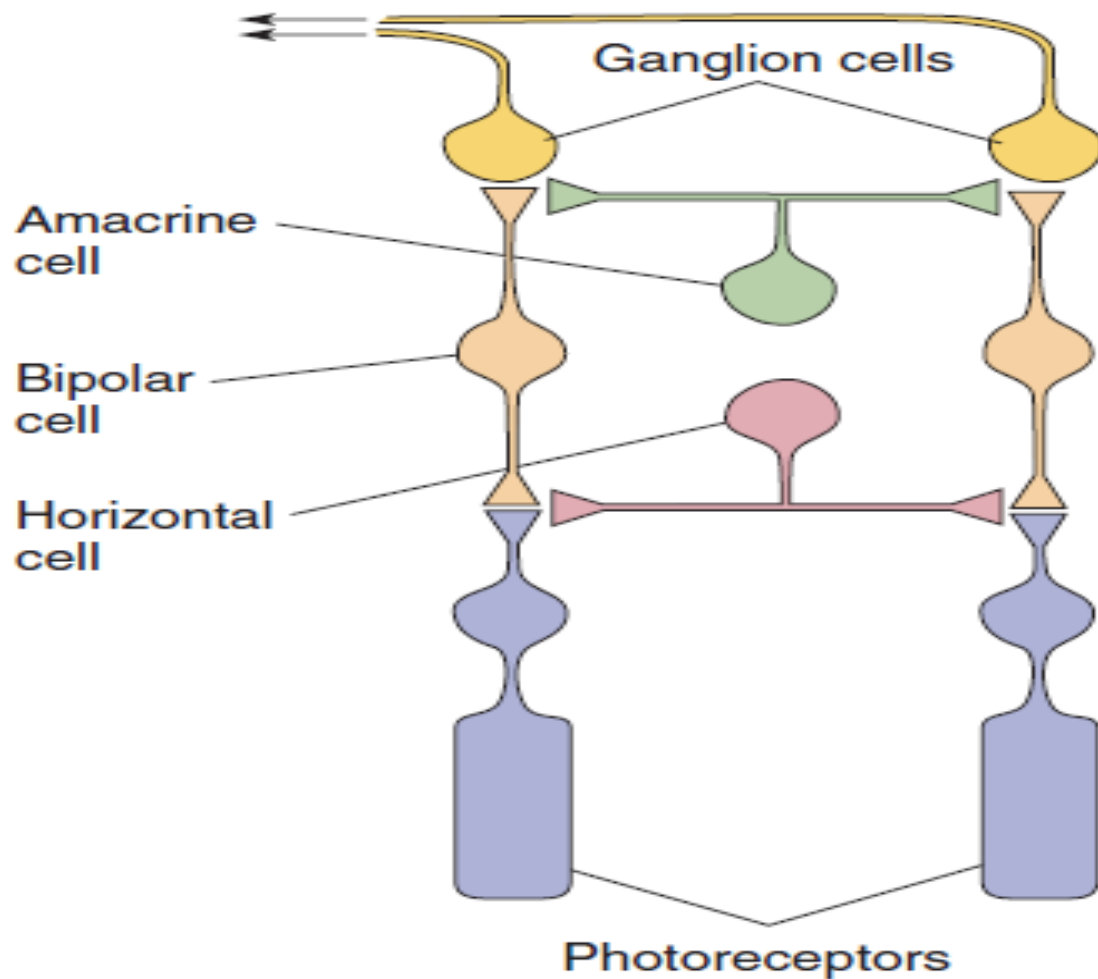
The Eye

- Vitreous Humor
 - Thick, gelatinous substance which fills the space between the back of the lens and the surface of the retina
 - Accounts $\approx 80\%$ of the volume of the eyeball
 - Maintain the shape of the eye
 - Contain **phagocytes** → remove blood and debris that might interfere with light transmission
 - *Floater* → too large debris particles which are difficult to be removed by phagocytes → cast shadows on the retina (in elderly)

The Retina

- Extends anteriorly almost to the ciliary body
- Contains **visual receptors** (rods and cones)
 - **Photoreceptors** → convert photons into electrical impulses
- Contain 4 types of neurons:
 - Bipolar cells
 - Ganglion cells
 - Horizontal cells
 - Amacrine cells

Ganglion cell axons
projecting to forebrain



Photoreceptors are the only
light-sensitive cells in the retina

Ganglion cells are the only
source of output from the retina

The most direct visual pathway:

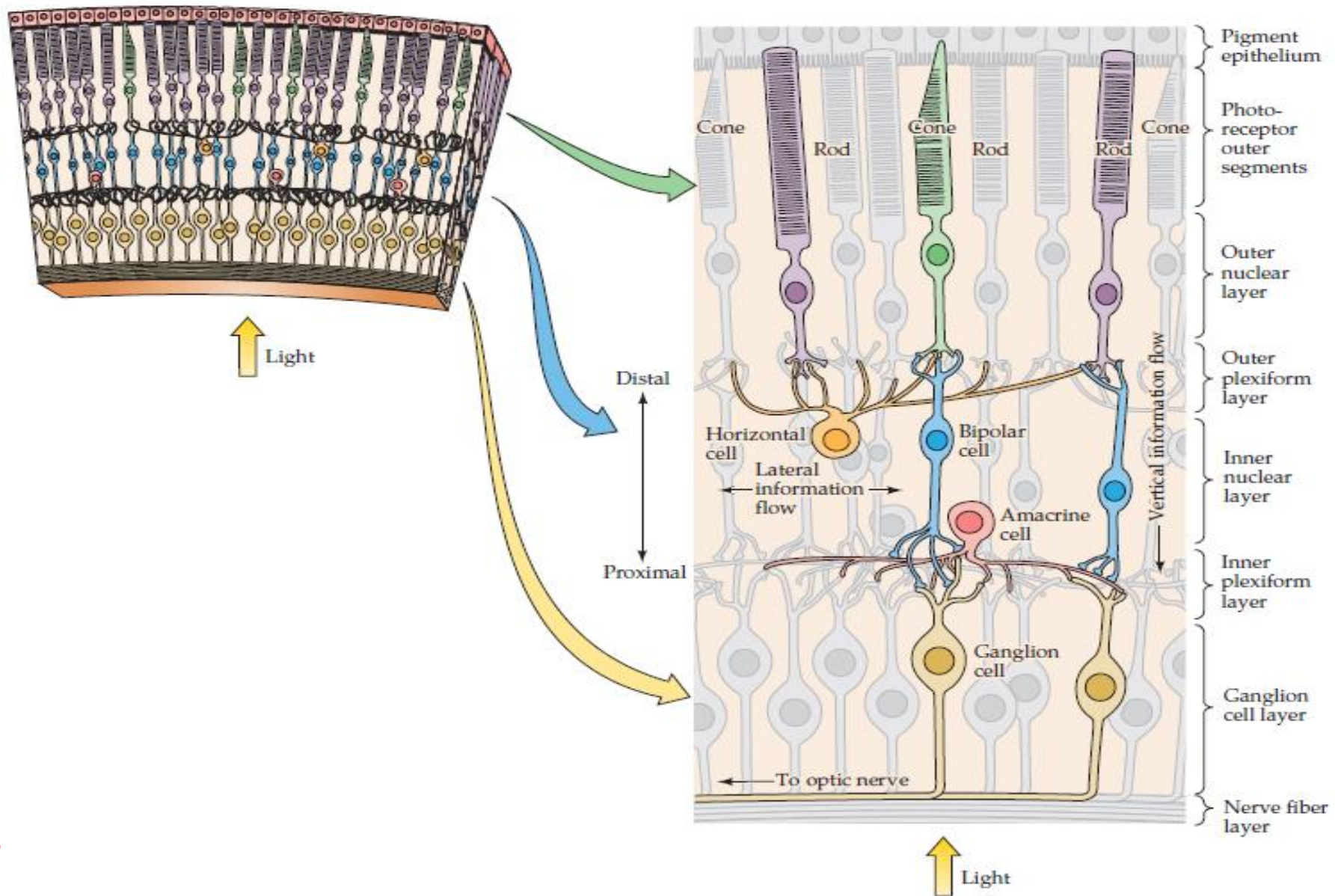
Photoreceptors → BCs → GCs

Retinal information processing

The Retina

- Laminar organization of the retina
 - Retinal cells are organized in layers inside-out
 - Light must pass from the vitreous humor through the GCs and BCs before it reaches the photoreceptors
 - Since retinal cells above photoreceptors are relatively transparent, image distortion is minimal as light passes through them

Layers of the Retina



The Retina

- Pigment epithelium layer:
 - Consists of pigmented (*melanin*) cuboidal cells:
 - Their bases are firmly attached to the choroidal layer located outside the retina
 - Tight junctions → prevent the flow of ions and plasma
 - Microvilli's (*at apical regions*) interdigitate with photoreceptors
 - Provide nutrition to the neural retina
 - Protect the retina from overexposure to light

The Retina

- Pigment epithelium layer:
 - **Melanin** absorb any light that is not captured by the retina and **reduces backscattering of light** to the retina (*otherwise may result in the degradation of the image*)
 - Plays a critical role in the maintenance of photoreceptors
 - **Bright light** causes pigment cells to close up to photoreceptors and retract when **light is dim**
 - Play a role in renewing photopigments and phagocytosing the photoreceptor disks (*their high turnover rate is essential to vision*)

Photoreceptors

- Rods and Cones:
 - Photosensitive cells (photoreceptors) of the retina
 - In most regions of the retina, **the rods outnumber the cones:**
 - $\approx 10^7$ rods & 5×10^6 cones
 - Have similar general structures and contain 3 segments:
 1. Outer segment
 2. Inner segment
 3. Synaptic terminal

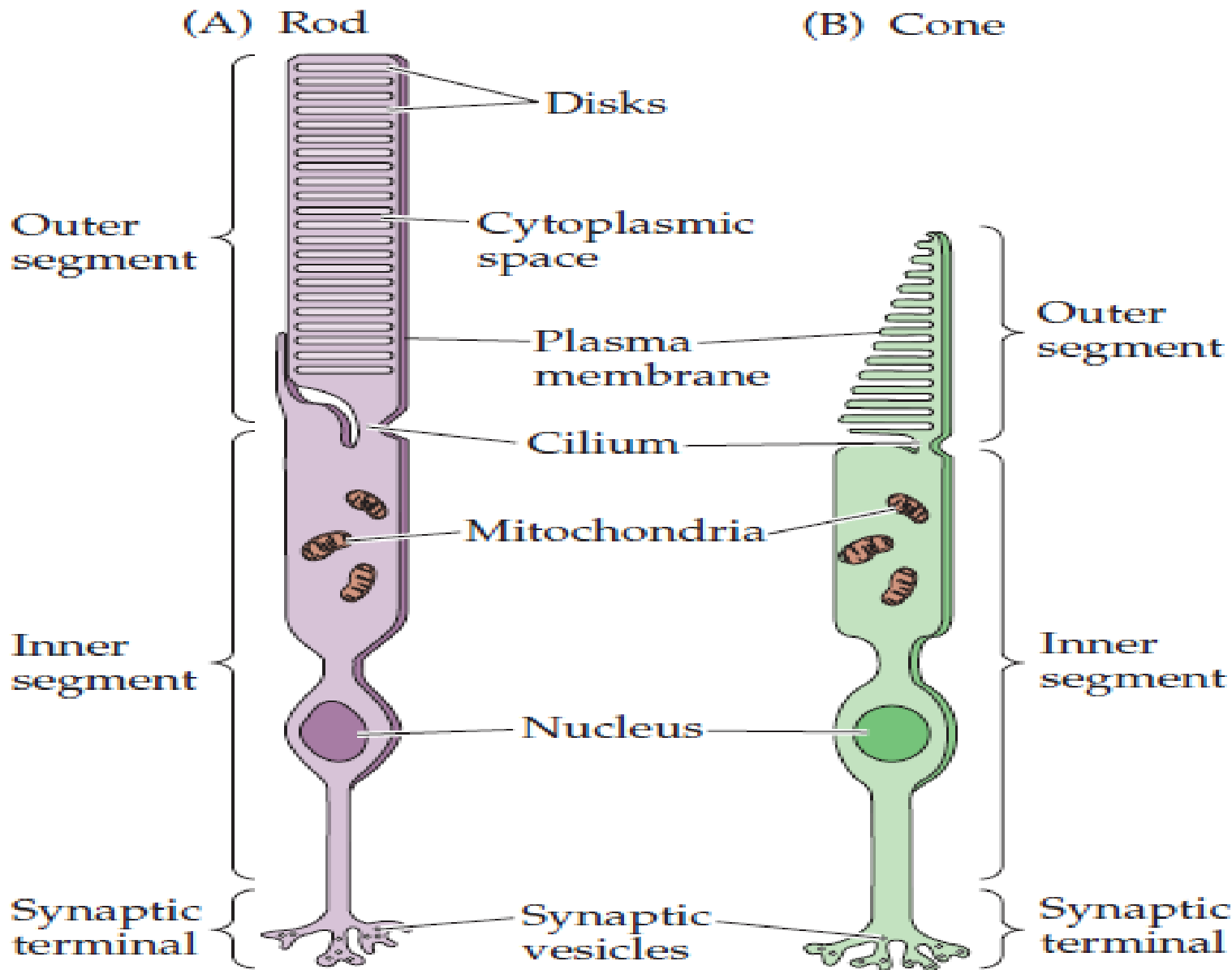
Photoreceptors

1. Outer segment

- Lies adjacent to the pigment epithelium
- Contain **light-sensitive photopigment** (*cone/rhodopsin*)
- Involved in **Phototransduction** (*convert photons to electrical signals*)
- Constantly renewed + discarded tips are removed by phagocytosis

2. Inner segment

- Contains the **cell nucleus** and most of the **biosynthetic mechanisms**
- Connected to the outer segment by **cilium** (*contains microtubules*)
- Gives rise to synaptic terminals



Photoreceptors

3. Synaptic terminal

- Signals from rods and cones are transmitted to the bipolar and horizontal cells via chemical synapses from presynaptic terminal
- Synaptic terminals of rods (*spherules*) and cones (*pedicles*) have sheets of proteins (*synaptic ribbons*)
- Synaptic ribbons function for **trafficking glutamate vesicles**
 - Photoreceptors release **glutamate tonically in the dark**
 - Glutamate release is **decreased in response to light**
- Makes synaptic contact with **bipolar and horizontal cells**

Photoreceptors

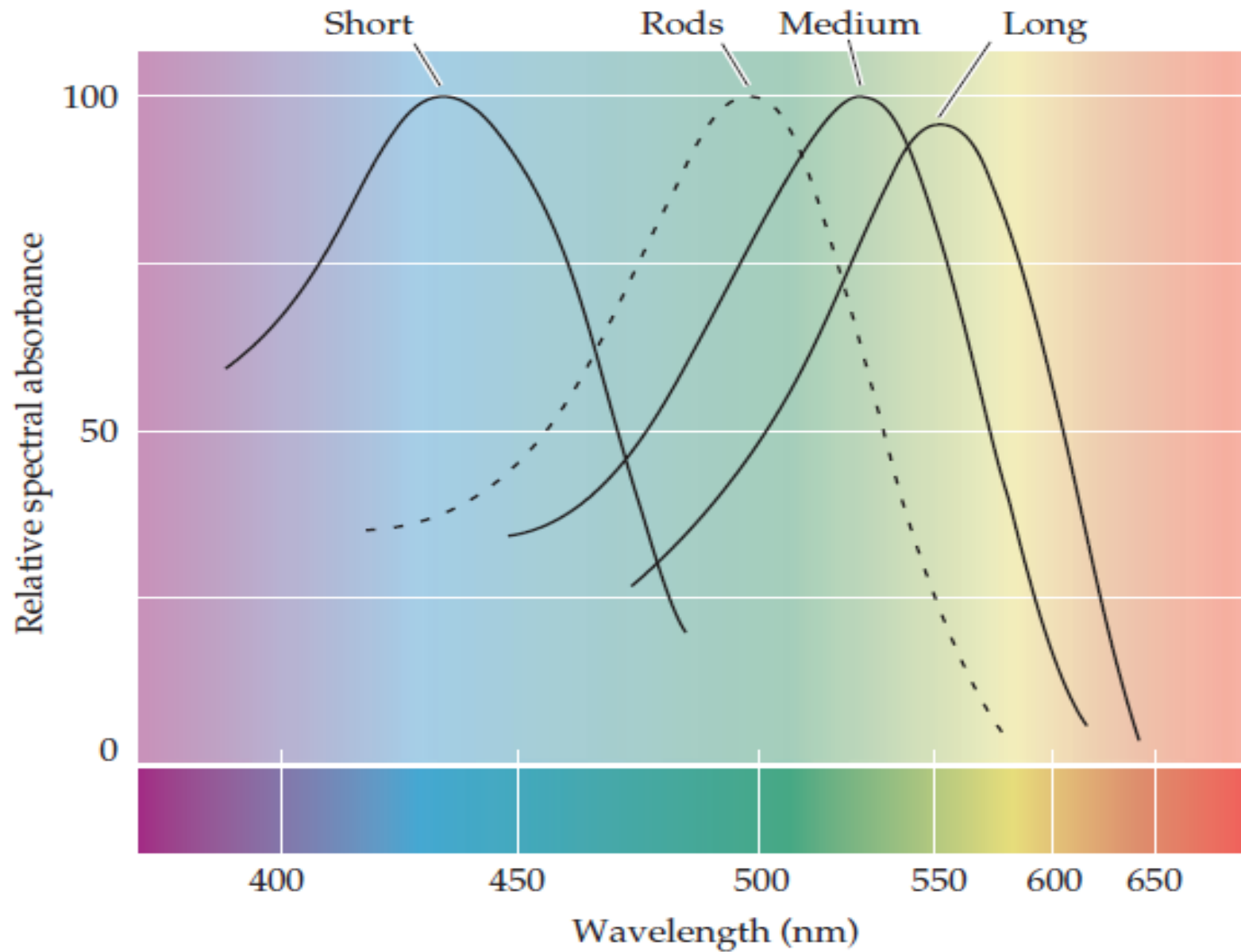
- Rods
 - Specialized for **night vision** (*saturate in daylight*)
 - Contain **more photosensitive pigment** than the cones which is responsible for the ability of rods to capture more light
 - **Highly sensitive** → can detect dim light ($\approx 1000\times$ than cones)
 - Have **slow response**, unlike cones
 - About half of the amino acid sequence of rhodopsin is identical with those of the three cone pigment proteins

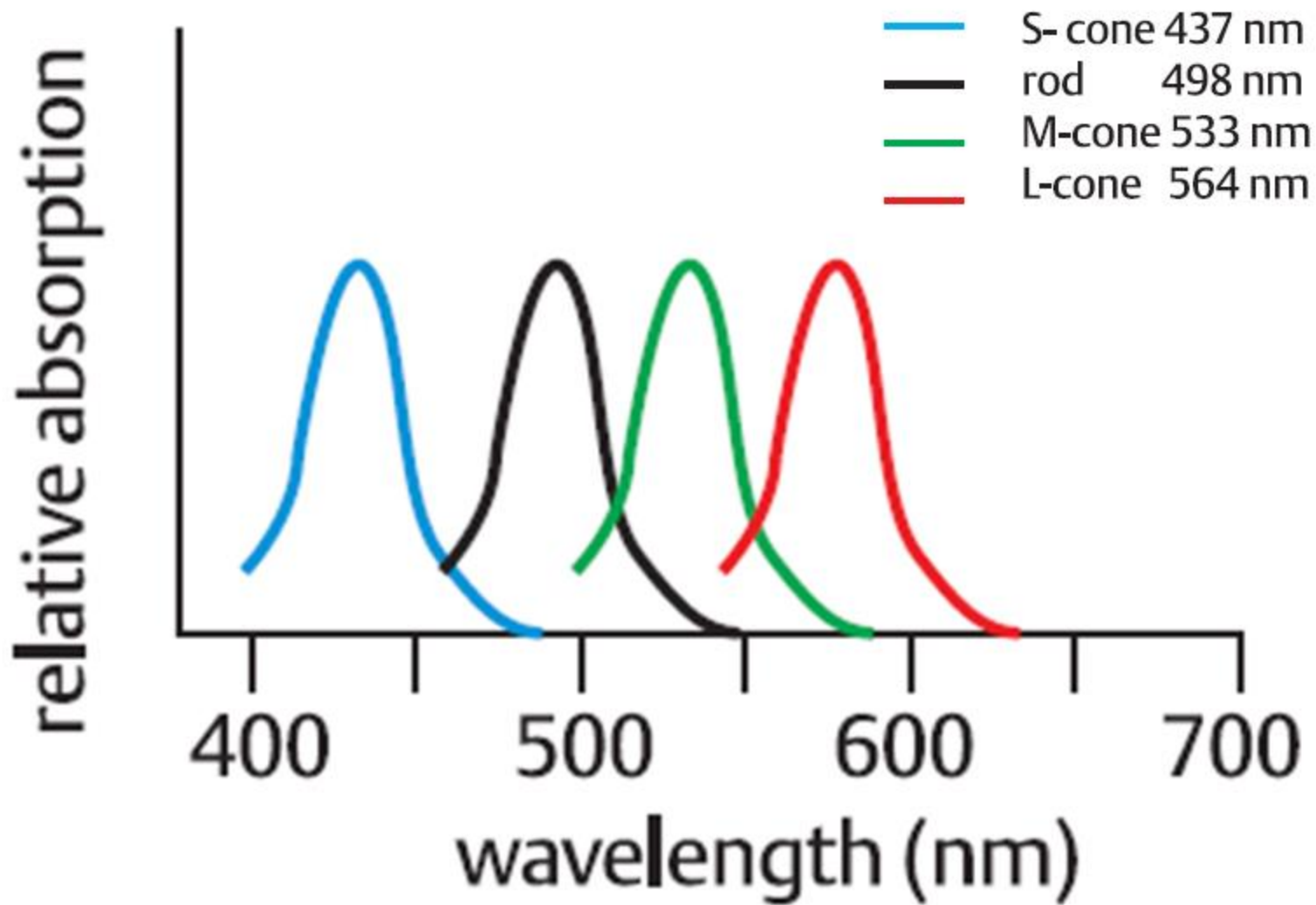
Photoreceptors

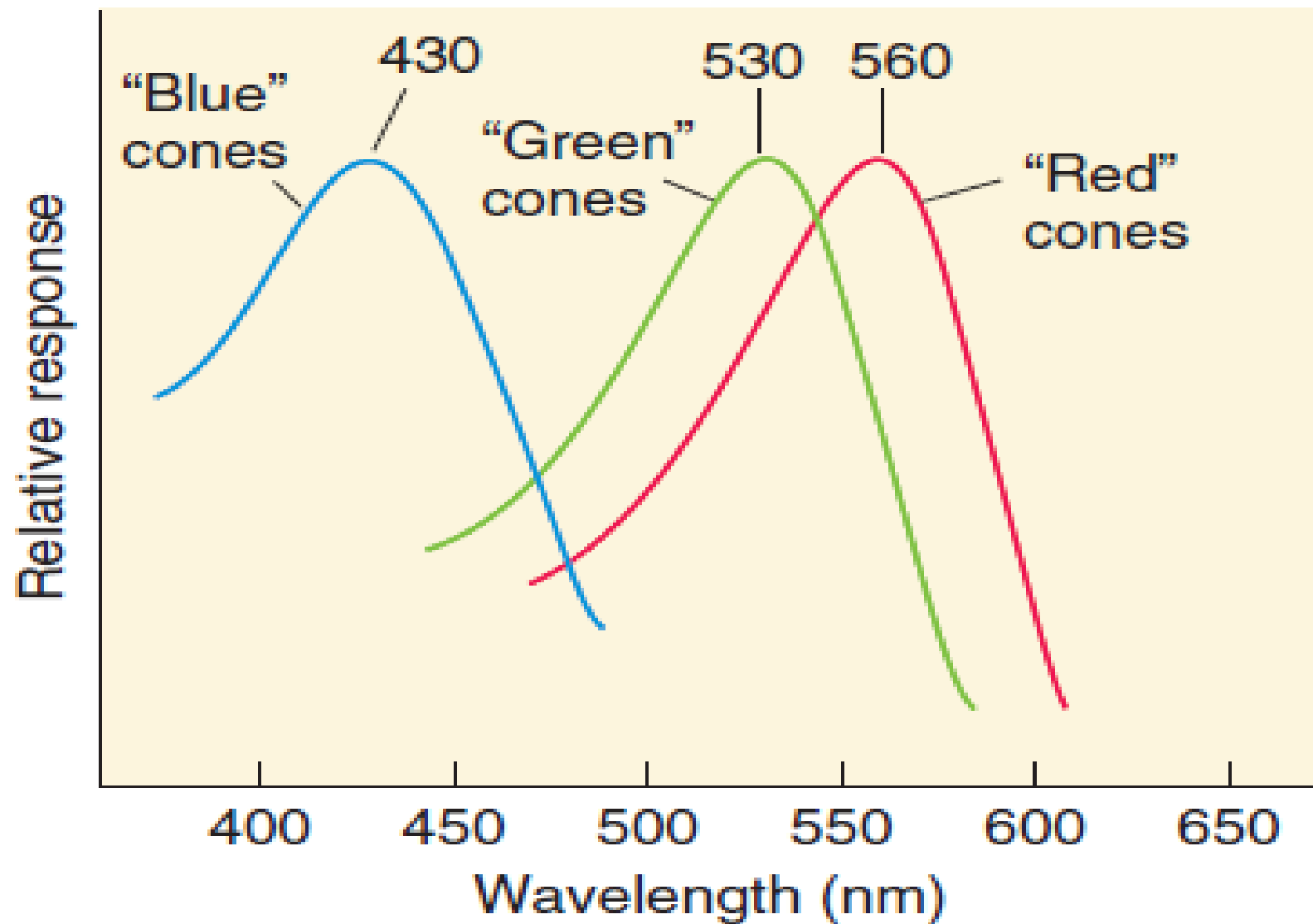
- Cones
 - Responsible for daylight vision
 - Mediate color vision (*unlike rods*)
 - Mediate higher visual acuity (*concentrated in the fovea*) than rods
 - Good visual acuity requires a low ratio of photoreceptors to ganglion cells
 - Their conical shape makes them more sensitive to direct axial rays
 - Have a fast response (short integration time)

Photoreceptors

- Cones:
 - 3 types of cones (with differing λ optima for absorption of light):
 - Each are excited maximally by different wavelengths of light
 - S-cones $\rightarrow$ absorb at shorter $\lambda \rightarrow$ Blue absorbing
 - M-cones $\rightarrow$ absorb at medium $\lambda \rightarrow$ Green absorbing
 - L-cones $\rightarrow$ absorb at longer $\lambda \rightarrow$ Red absorbing
 - Green & Red absorbing proteins are X-chromosome linked (*differ in only 15 of their 348 residues*)
 - Blue absorbing proteins are encoded on an autosomes



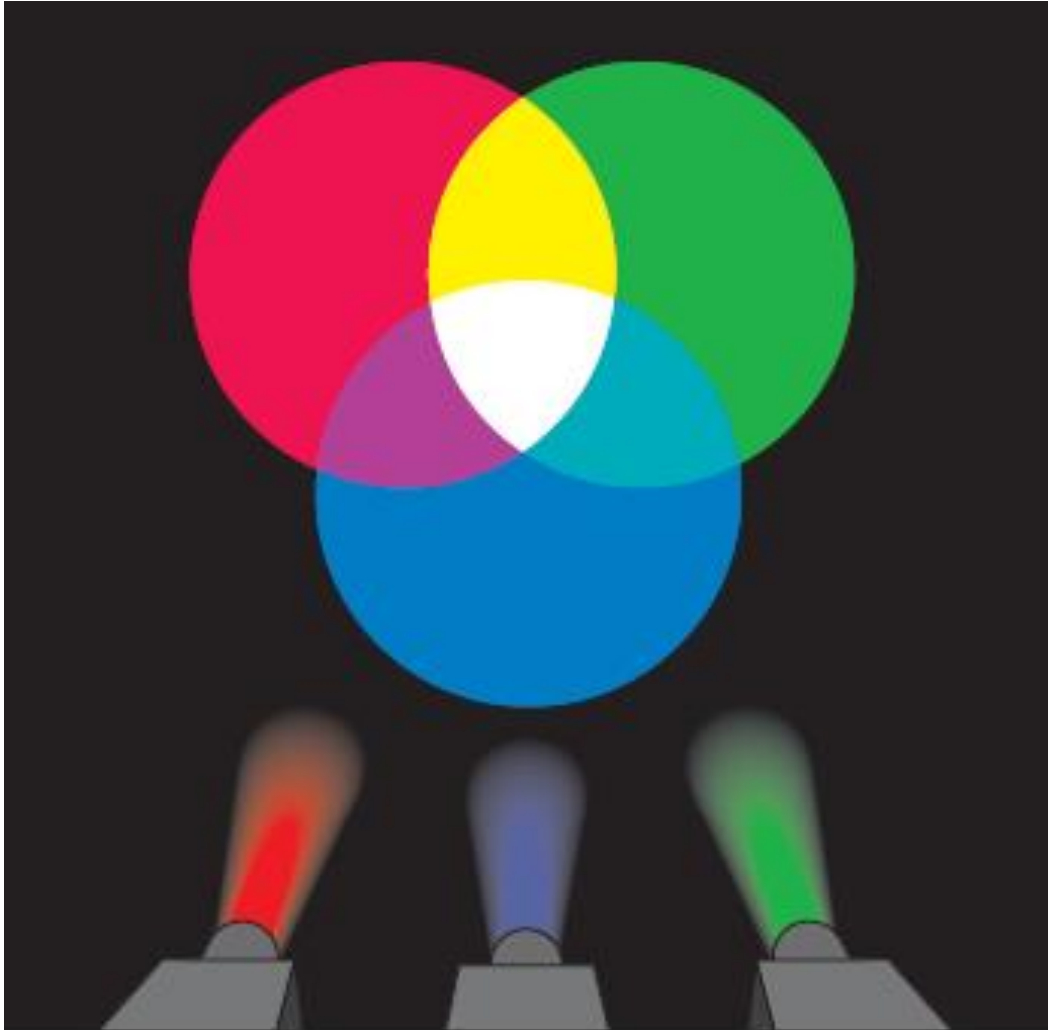




Photoreceptors

- Cones & color vision
 - Perception of color is largely determined by the relative contributions of *blue, green, and red cones* to the retinal signal
 - All the colors of the rainbow are created by mixing the proper ratio of red, green, and blue light
 - The brain assigns colors based on a comparison of the readout of the three cone types
 - **White color** is perceived when all types of cones are equally active
 - The peak sensitivity of *rods is $\lambda \approx 500\text{nm}$* (perceived as *blue-green*)

Mixing colored lights



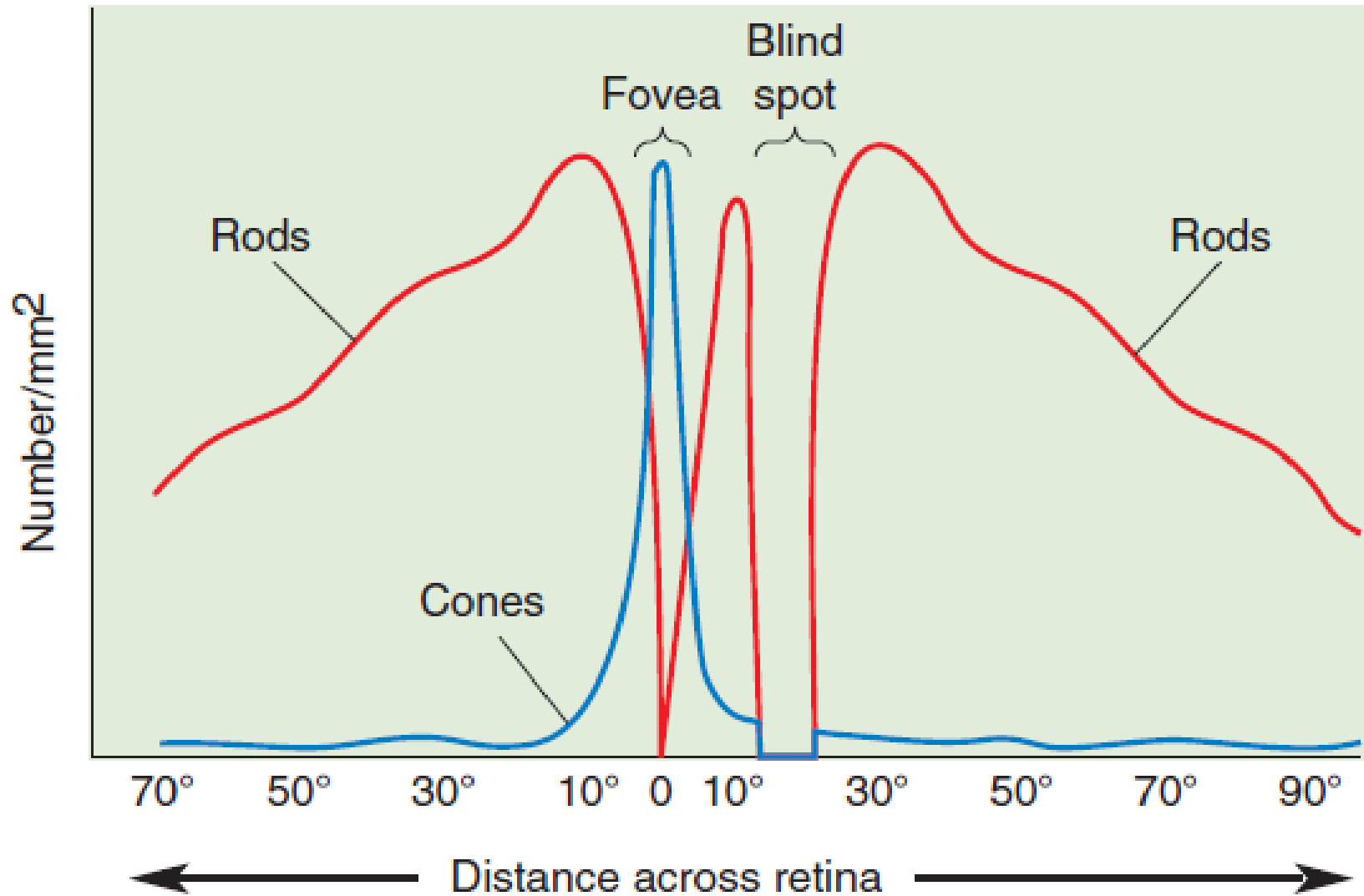
White perception results when mixing of red, green, and blue light causes equal activation of the 3 types of cones

Photoreceptors

- Rods & Cones: Retinal distribution

1. Peripheral retina

- Higher ratio of rods to cones
 - *Rods are specialized for low light*
- Higher ratio of photoreceptors to ganglion cells
 - *More photoreceptors feeding information to each GC*
- Thus, the peripheral retina is **more sensitive to light**

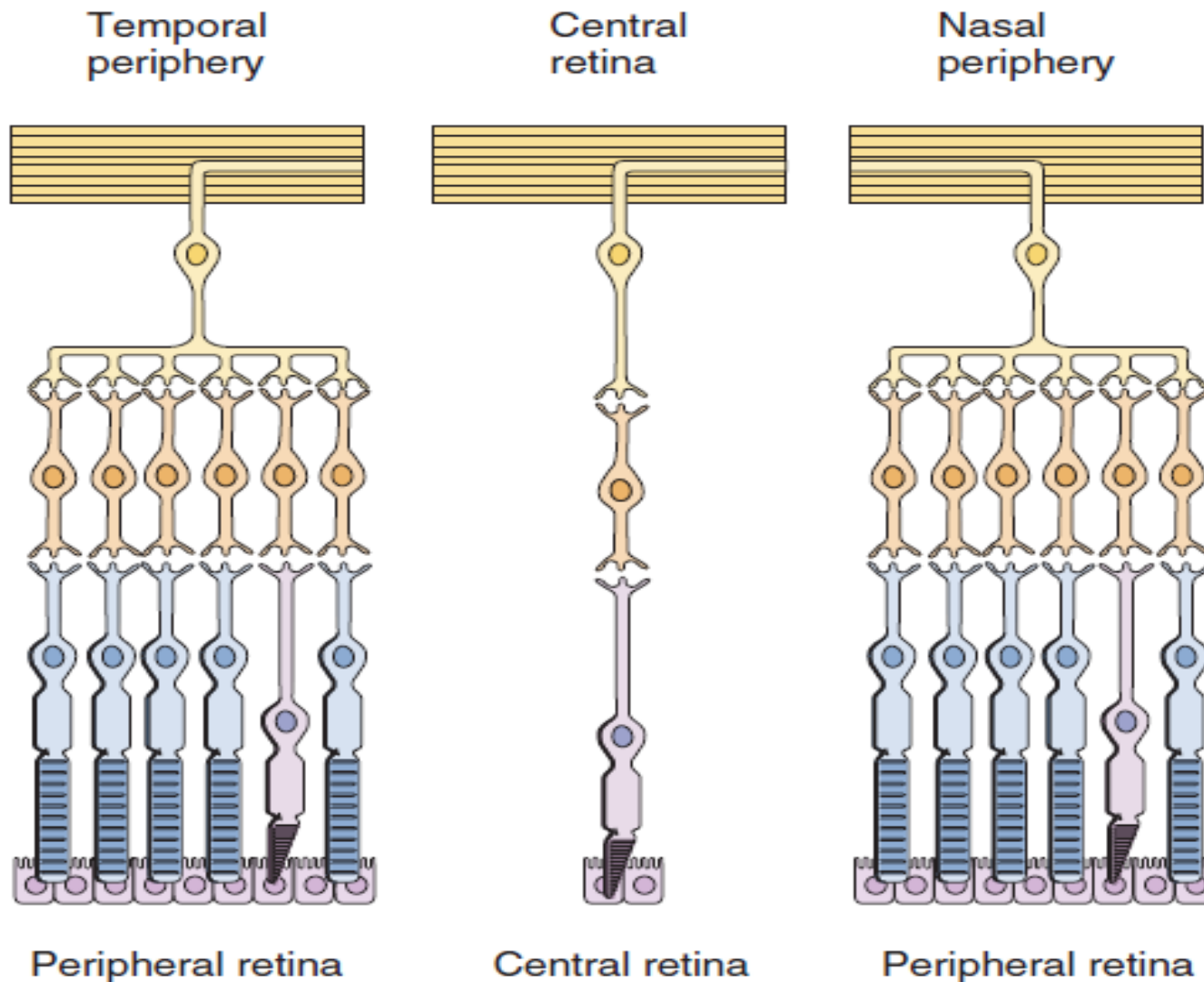


Regional differences of retinal

Photoreceptors

2. Central retina

- Contain Cones, within 10° of the fovea
 - *Rods are absent from the central fovea*
- Relatively few photoreceptors feed information directly to GCs
- Thus, the central retina is better for **high-resolution vision**
 - Good visual acuity requires a low ratio of photoreceptors to GCs



Regional differences of retinal

Photoreceptors

- **NB:**
 - Loss of cones results in decreased **visual acuity + color vision deficits**
 - **Color blindness** is caused by sex-linked mutations or deletions of genes encoding cone pigments
 - *Individuals with loss of cones are considered to be legally blind*
 - Loss of rods results in *night blindness + loss of peripheral vision*

The Physiology of Vision

Image Formation

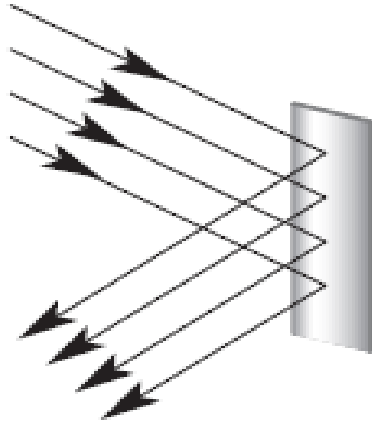
Image Formation

- Refraction of light:
 - Light rays travel through air at a velocity of $\approx 300,000$ km/sec
 - Light travel much slower through transparent solids and liquids
 - **Refractive index** $\rightarrow$ ratio of velocity of light in air to the velocity in the substance
 - Light rays passing through **perpendicular interface** $\rightarrow$ enter the second medium without deviation (slow & shorter λ)
 - Light rays passing through **angulated interface** $\rightarrow$ light rays bend if the refractive indices of the two media are different

Image Formation

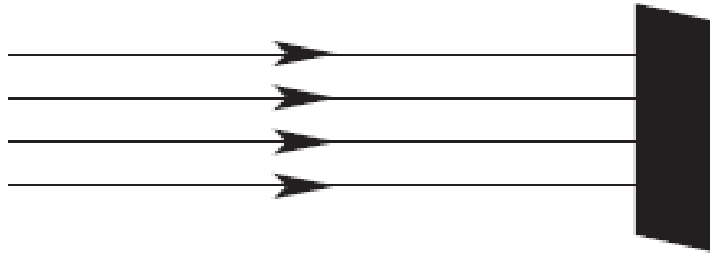
- Refraction of light:
 - Degree of refraction depends on the:
 - Ratio of the refractive indices of the two transparent media
 - Degree of angulation between the interface and the entering wave front (curvature of the interface)
 - Curvature of cornea is constant, but the curvature of the lens can be varied (*fine control for focusing light on the retina*)
 - As a result of light refraction, the image formed on the retina is upside down and right to left

Reflection

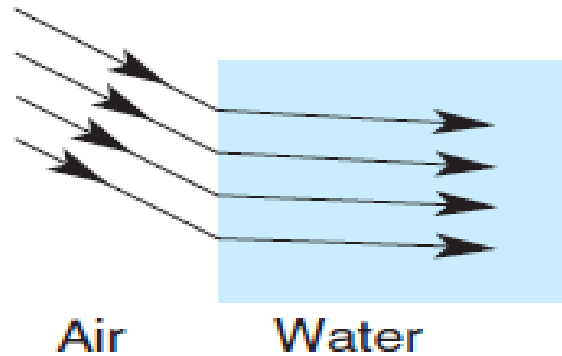


Reflection and absorption determine
what light enters the eye

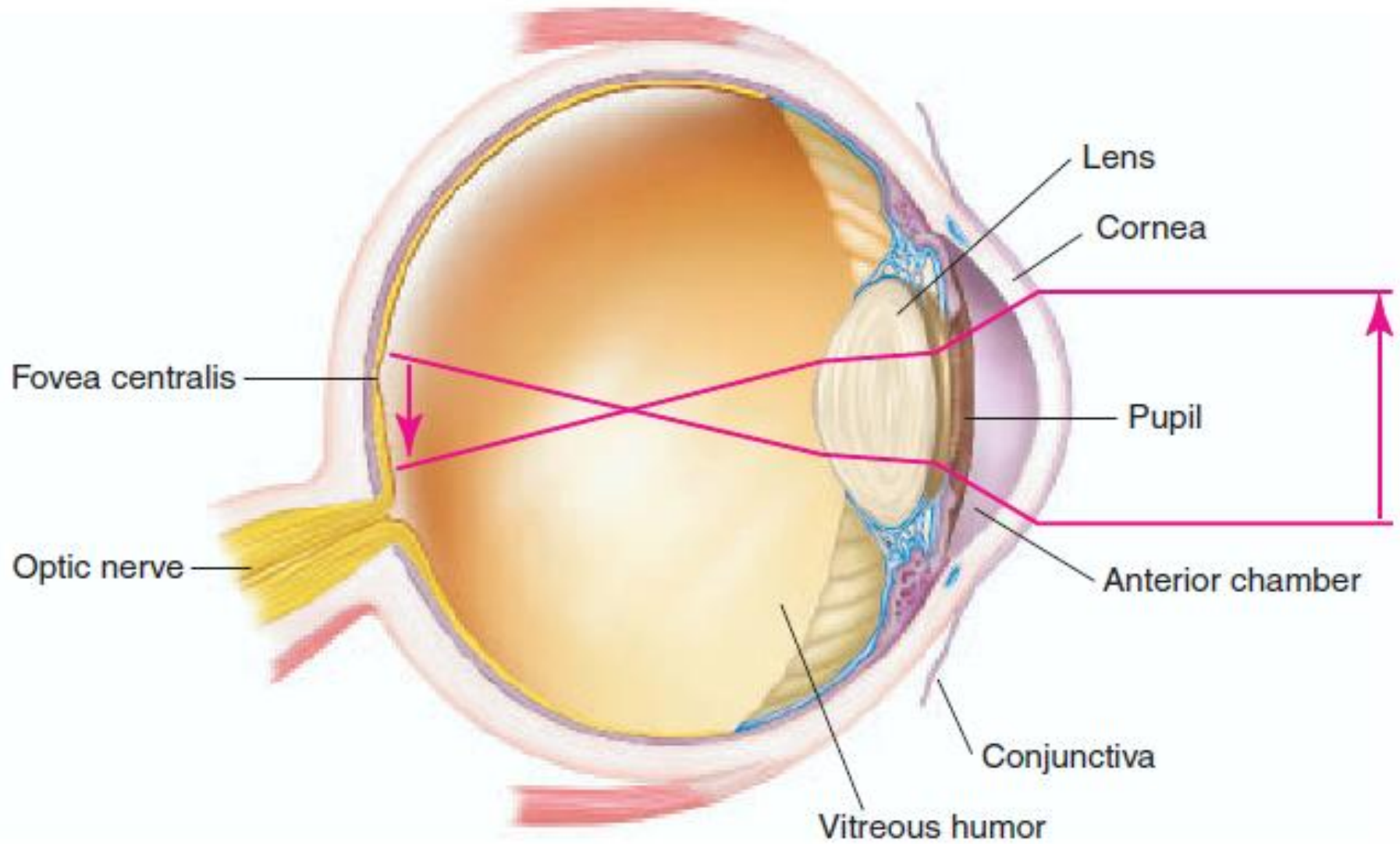
Absorption



Refraction



Interaction of Light with the Environment



Light refraction by the Cornea

Image Formation

- Accommodation:
 - When a normal eye views an object, parallel rays of light are refracted to a point, or focus, on the retina
 - Light rays originating from near point diverge and require greater refractive power to focus on the retina
 - This additional focusing power is provided by changing the shape of the lens (*Accommodation*)
 - *Distant objects* require little refraction provided by a *flat lens*
 - *Near objects* require greater refraction provided by a more *rounded lens*

Image Formation

- Accommodation:
 - Viewing an object placed 20 feet away or more
 - Ciliary muscle relax → wide aperture → ↑ tension in the suspensory ligaments → the lens is stretched into a flatter shape → ↓curvature of the lens → ↓refractive power
 - As the object moves closer to the eyes:
 - Ciliary muscle contract → narrow aperture → ↓ tension on the suspensory ligaments → the lens becomes rounder and thicker → ↑curvature of the lens → ↑refractive power → Accommodation

Far point

Flat lens

Near point

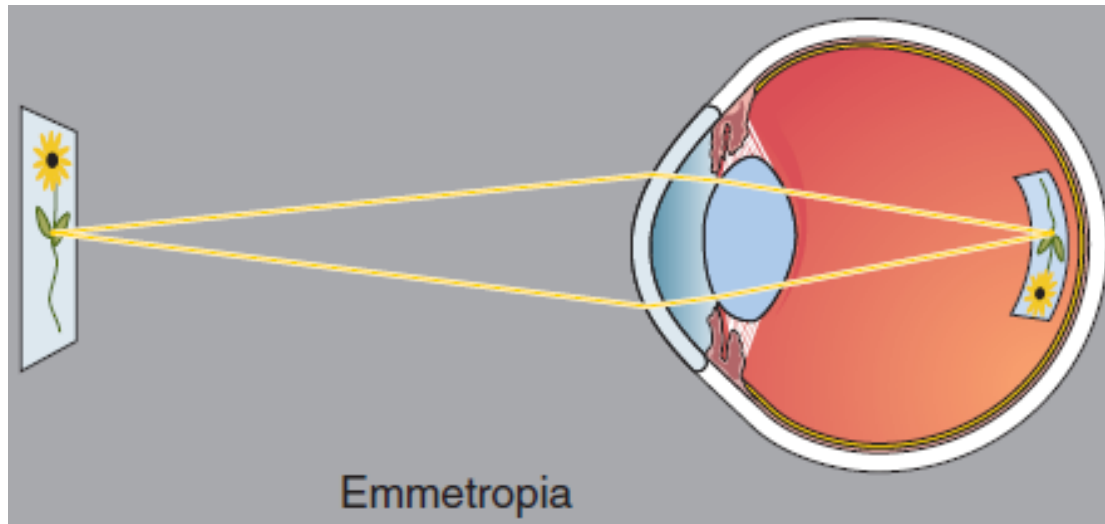
Fat lens

Accommodation by the Lens

Vision Correction

1. Emmetropia:

- Sharply focusing of parallel light rays from a distant point source onto the retina (*when the ciliary muscles are relaxed and the lens is flat*)
- i.e. emmetropic eye focuses parallel light rays on the retina without the need for accommodation (*normal vision*)



Vision Correction

2. Hyperopia (farsightedness):

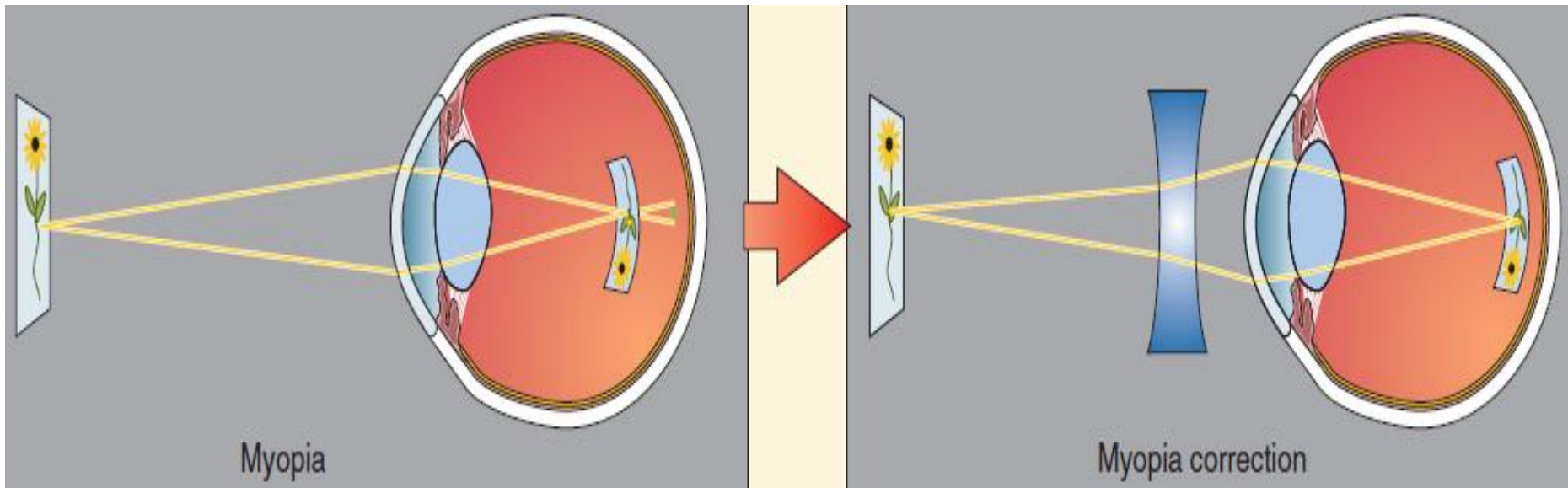
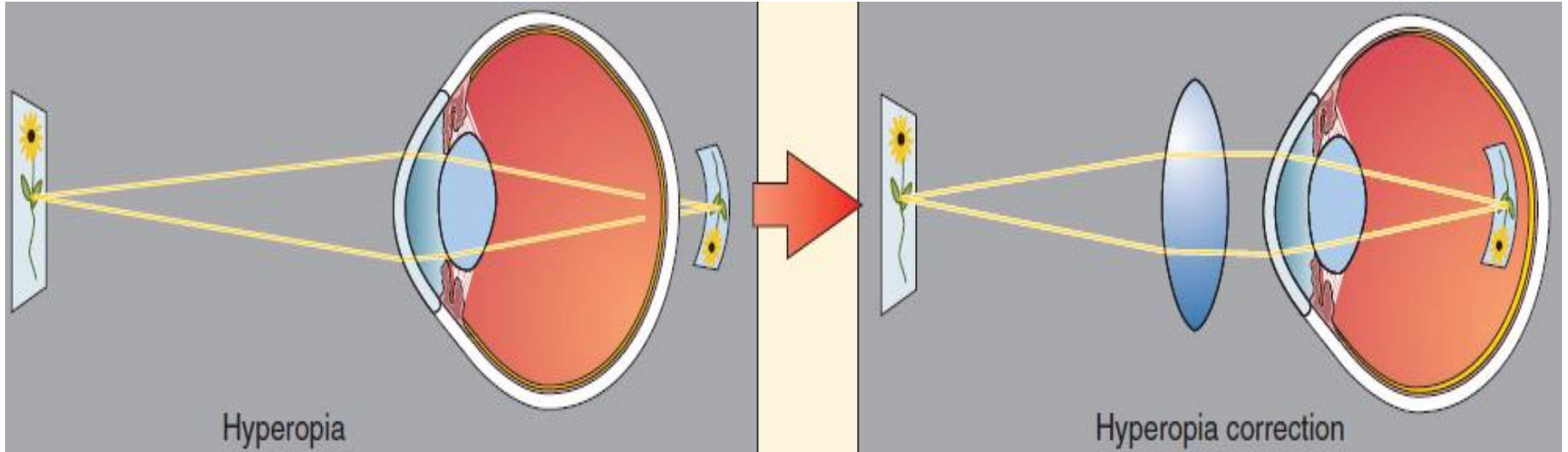
- When the eyeball is too short from front to back
- The eye can focus on far objects but the lens cannot accommodate enough to form an image on near points
- Light rays are focused **behind the retina** → Corrected by a **convex lens**
- The curved front edge of the lens (*like the cornea*) bends light toward the center of the retina

Vision Correction

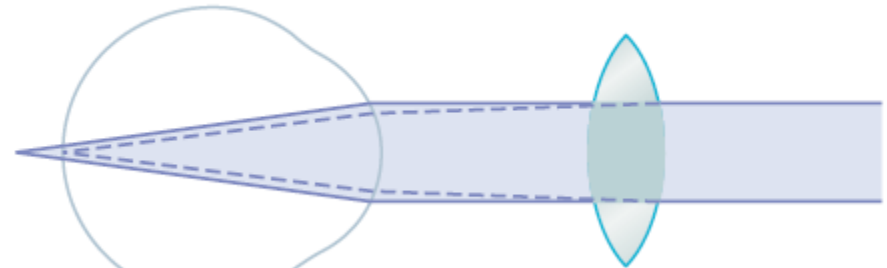
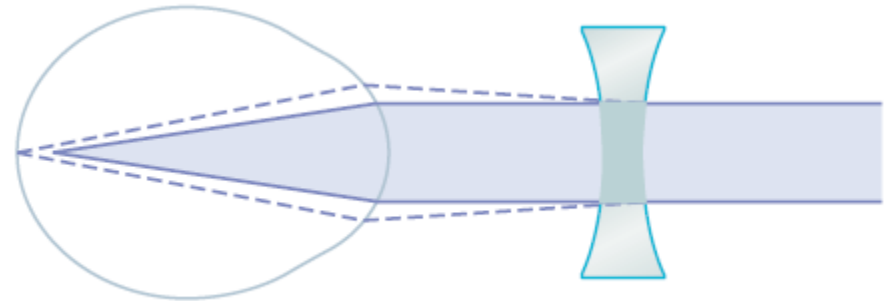
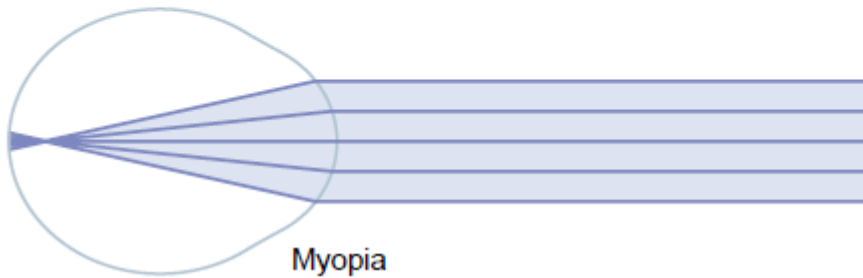
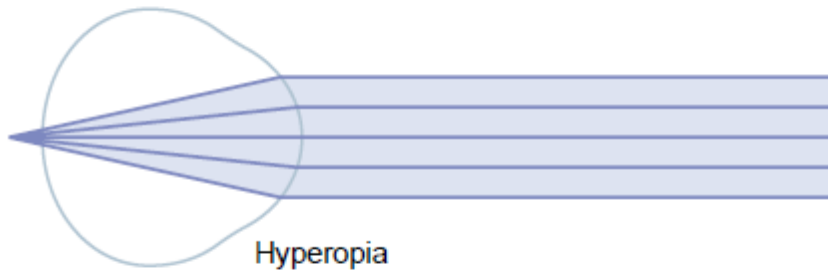
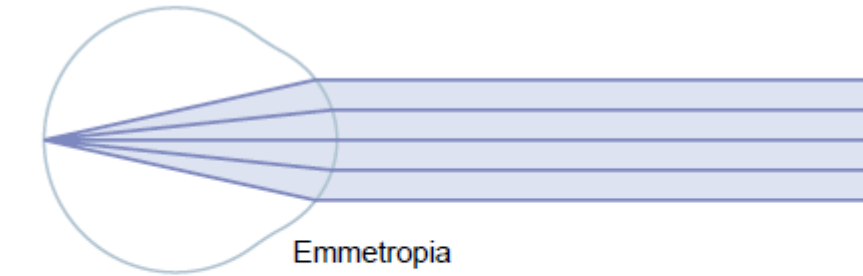
3. Myopia (nearsightedness):

- When the eyeball is too long (*stretched*)
- The eye can focus on near objects
- The amount of refraction provided by the cornea and the lens is too great to focus distant objects
- Light rays will converge **before the retina**
- Corrected by a **concave lens** → **diverges light rays**, so that the point of focus is farther from the lens and is thus pushed back to the retina

Vision Correction



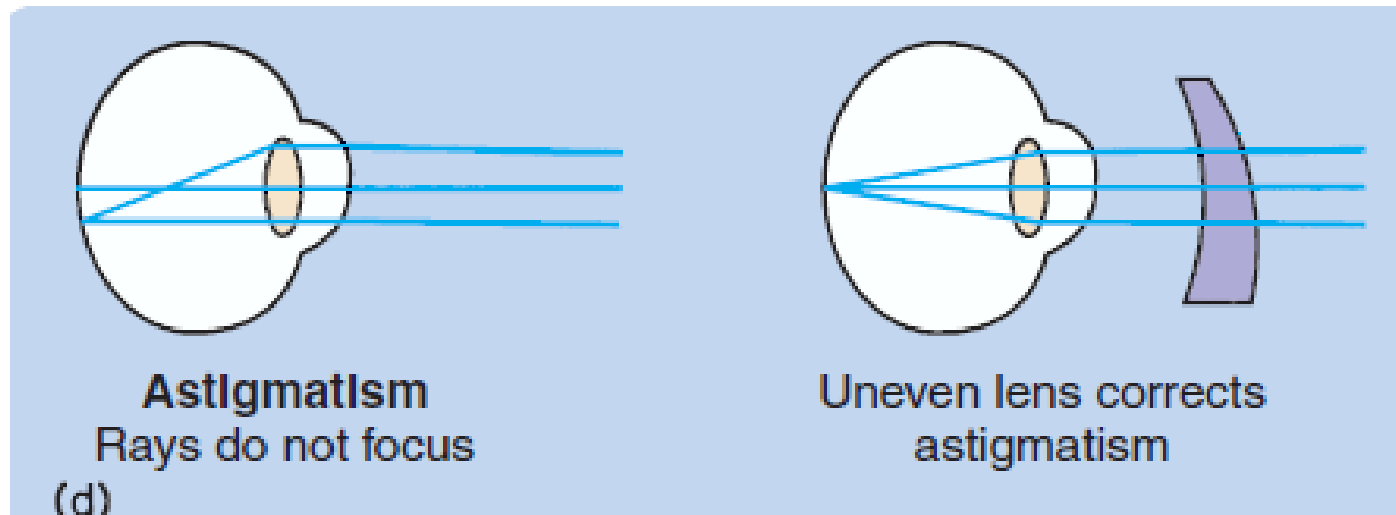
Vision Correction



Vision Correction

4. Astigmatism:

- When the **asymmetry** of the cornea and/or lens is significant
- Corrected by **cylindrical lenses** that compensate for the asymmetry in the cornea or lens of the eye



Physiology of Vision

Phototransduction

Phototransduction

- Transduction of light energy into electrical signals through a primary photoelectric effect on the photopigment
 - **Photopigment** → Rhodopsin + Cone-pigment
- **Occurs the same way both in cones and rods**
 - The only major difference is in the **type of opsins** in the membranous disks of the cone outer segments
- Photoreceptors respond to light with graded changes in membrane potential (*not with an action potential*)

Phototransduction

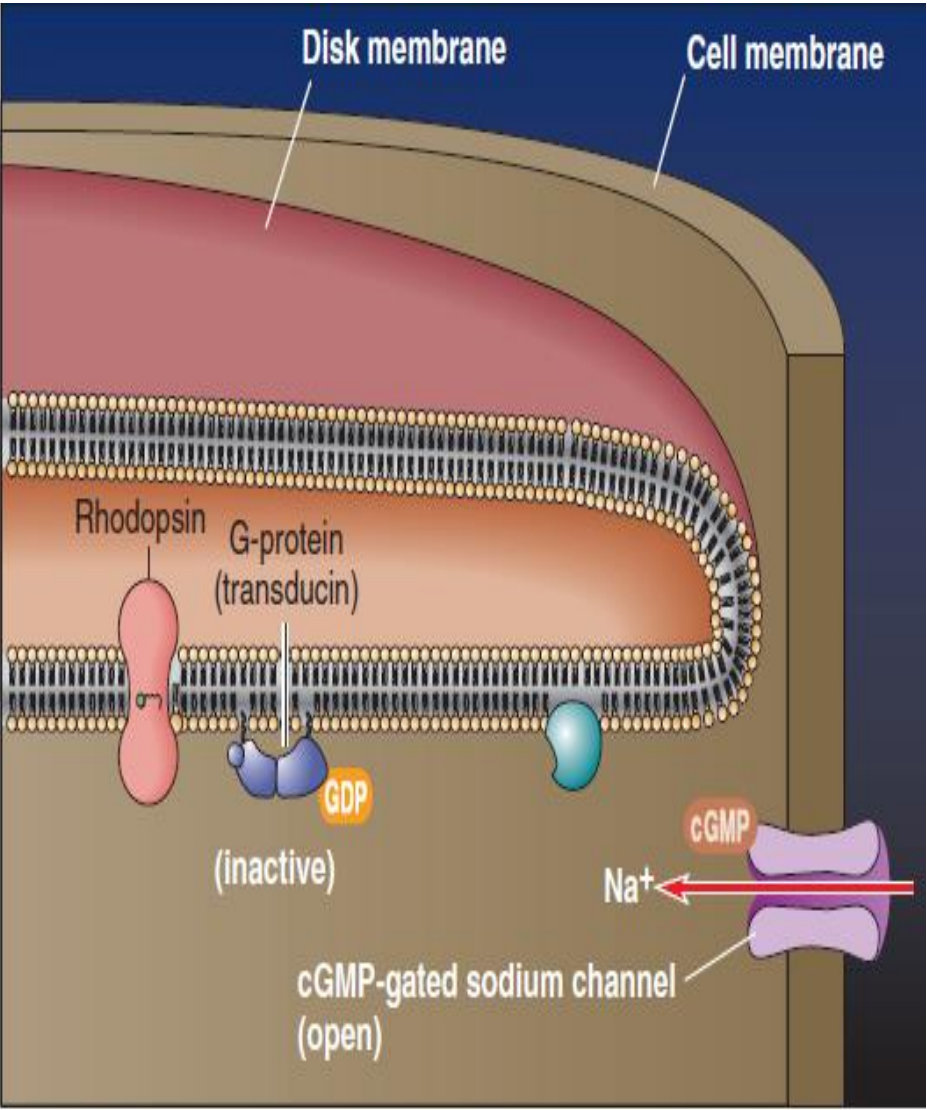
- Photopigment:
 - Consists of opsin attached with a chromophore (retinal)
 - Opsin → a disaccharide-protein complex + do not absorb light
 - Chromophore → the part of a molecule that absorbs light
 - Retinal → an aldehyde form of vitamin-A (*11-trans retinol*)
 - Generated by the pigment epithelium as *11-cis retinal* + transported by a retinal-binding protein
 - 11-trans retinol cannot be synthesized in our body

Phototransduction

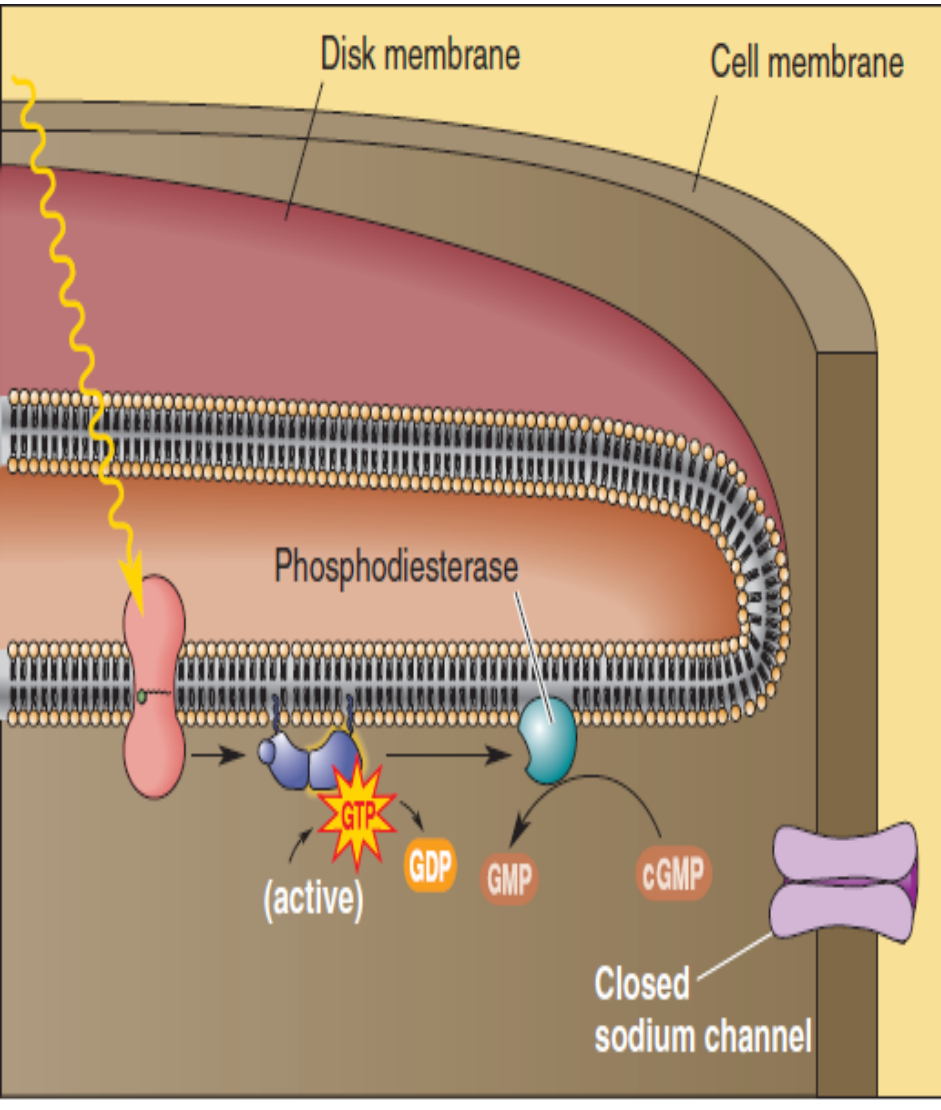
- Photopigment:
 - Absorption of light changes conformation of *11 cis-retinal* to *all-trans retinal*) causing it to **dissociate from the opsin** → **Bleaching**
 - This dissociation reaction in response to light initiates changes in the ionic permeability of the rhodopsin membrane
 - 11-cis retinal is then slowly regenerated in the pigment epithelium (*reason for the slow rate of dark adaptation*)
 - Bleaching stimulates a G-protein (*Transducin*) in the disk membrane, which in turn **activates PDE**

Phototransduction

- Polarization of photoreceptors:
 - In complete darkness, membrane potential of the outer segment is depolarized to $\approx -30\text{mV}$ due to inward Na^+ current (*dark current*)
 - Influx of Na^+ is mediated through cGMP-gated Na^+ channels located in the outer segment of photoreceptors
 - cGMP binds to the cytoplasmic side of the channel
 - $[\text{cGMP}]_i$ increases during darkness and decreases during light
 - $\uparrow[\text{cGMP}]_i$ opens Na^+ channels $\rightarrow$ influx of $\text{Na}^+ / \text{Ca}^{2+}$
 - $\downarrow [\text{cGMP}]_i$ leads to closure of Na^+ channels



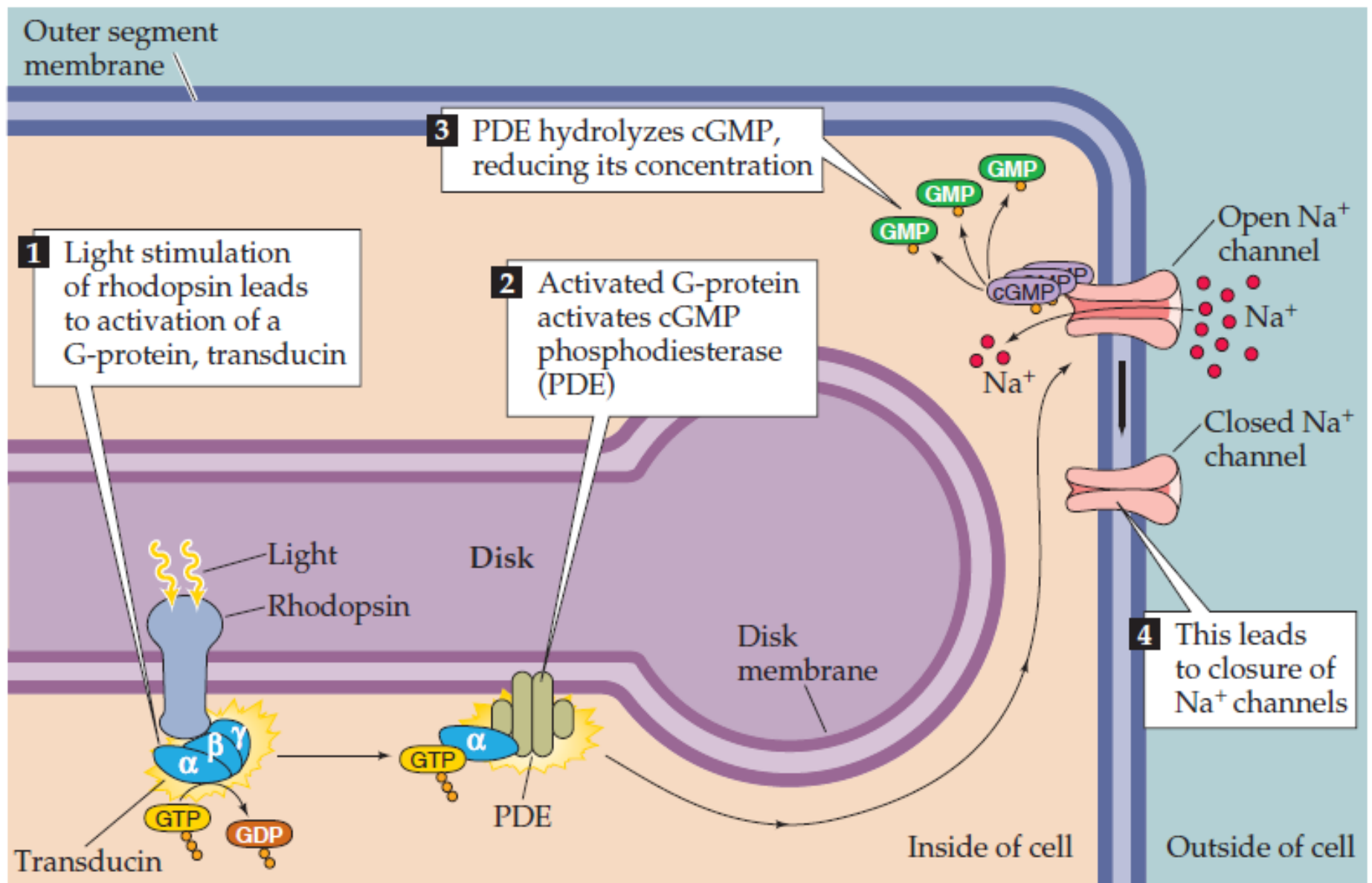
(a) Dark

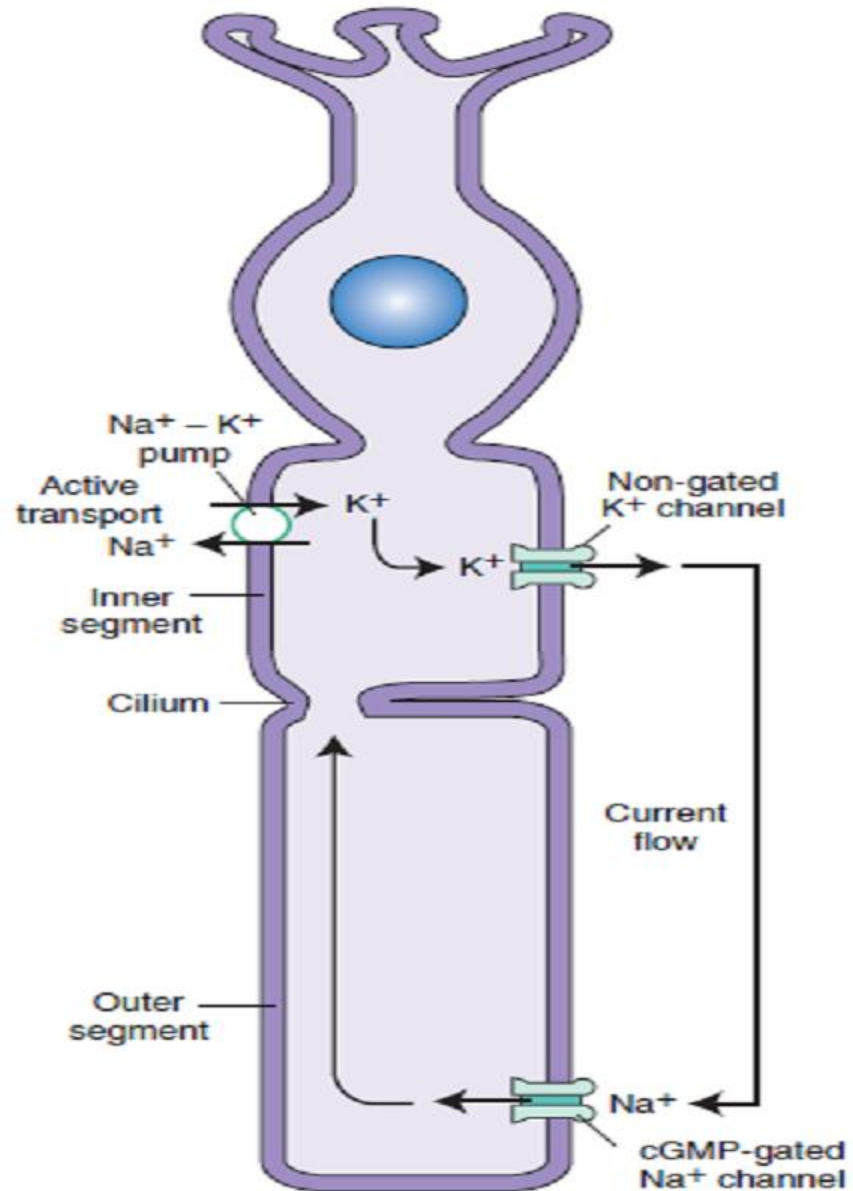
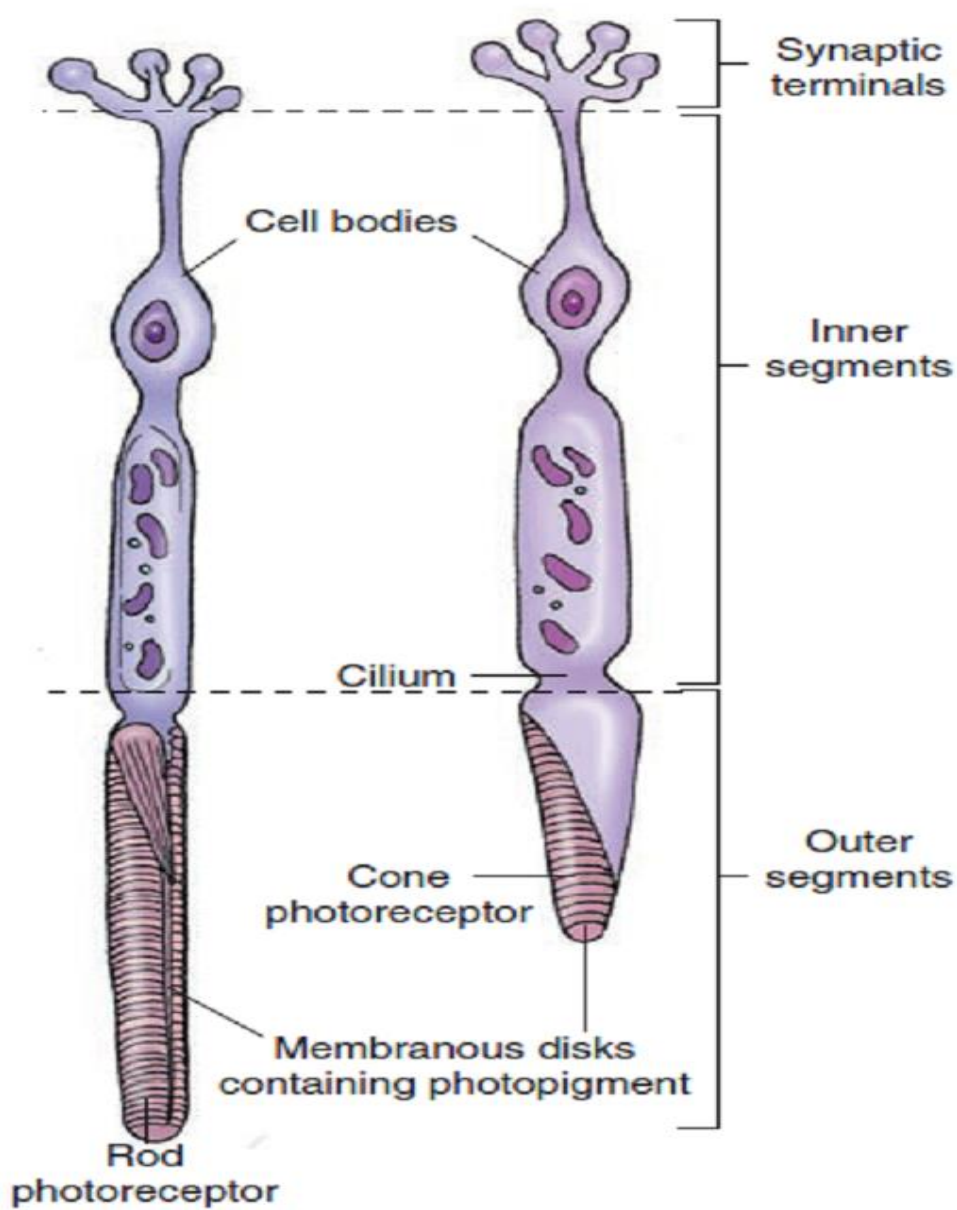


(b) Light

Phototransduction

- During Dark:
 - $\uparrow[\text{cGMP}] \rightarrow$ opening of Na^+ channels $\rightarrow$ influx of $\text{Na}^+ \rightarrow$ **depolarization** of photoreceptors
 - **$\text{Na}^+ - \text{K}^+$ pump** maintains excitability
 - Efflux of K^+ across the inner segment through leaky K^+ channels
- During Light:
 - Light stimulation of rhodopsin $\rightarrow$ activate G-protein (*Transducin*)
 $\rightarrow$ activate PDE $\rightarrow$ hydrolyze cGMP $\rightarrow \downarrow[\text{cGMP}]_i \rightarrow$ closure of Na^+ channels $\rightarrow$ **hyperpolarization** of photoreceptors





Phototransduction

- Dark adaptation:
 - Transition from all-cone daytime vision to all-rod nighttime vision
 - Not instantaneous → takes about 20–25 minutes
 - Sensitivity to light increases a million fold during darkness
 - Mechanisms:
 - Dilation of the pupils, which allows more light to enter the eye
 - Regeneration of unbleached rhodopsin
 - Adjustment of the functional circuitry of the retina → information from more rods is available to each ganglion cell

Phototransduction

- Light adaptation:
 - Ability of the eye to adapt to changes in light level
 - Transition from all-rod nighttime vision to all-cone daytime vision
 - Reversing changes in the retina that accompanied dark adaptation
 - Takes about 5–10 minutes
 - Constriction of the pupil reduce the light entering the eye
 - Transition of dark-adapted eye into bright light temporarily saturates photoreceptors (due to ↑light sensitivity)
 - Sensitivity of the eye to light is correspondingly reduced

Physiology of Vision

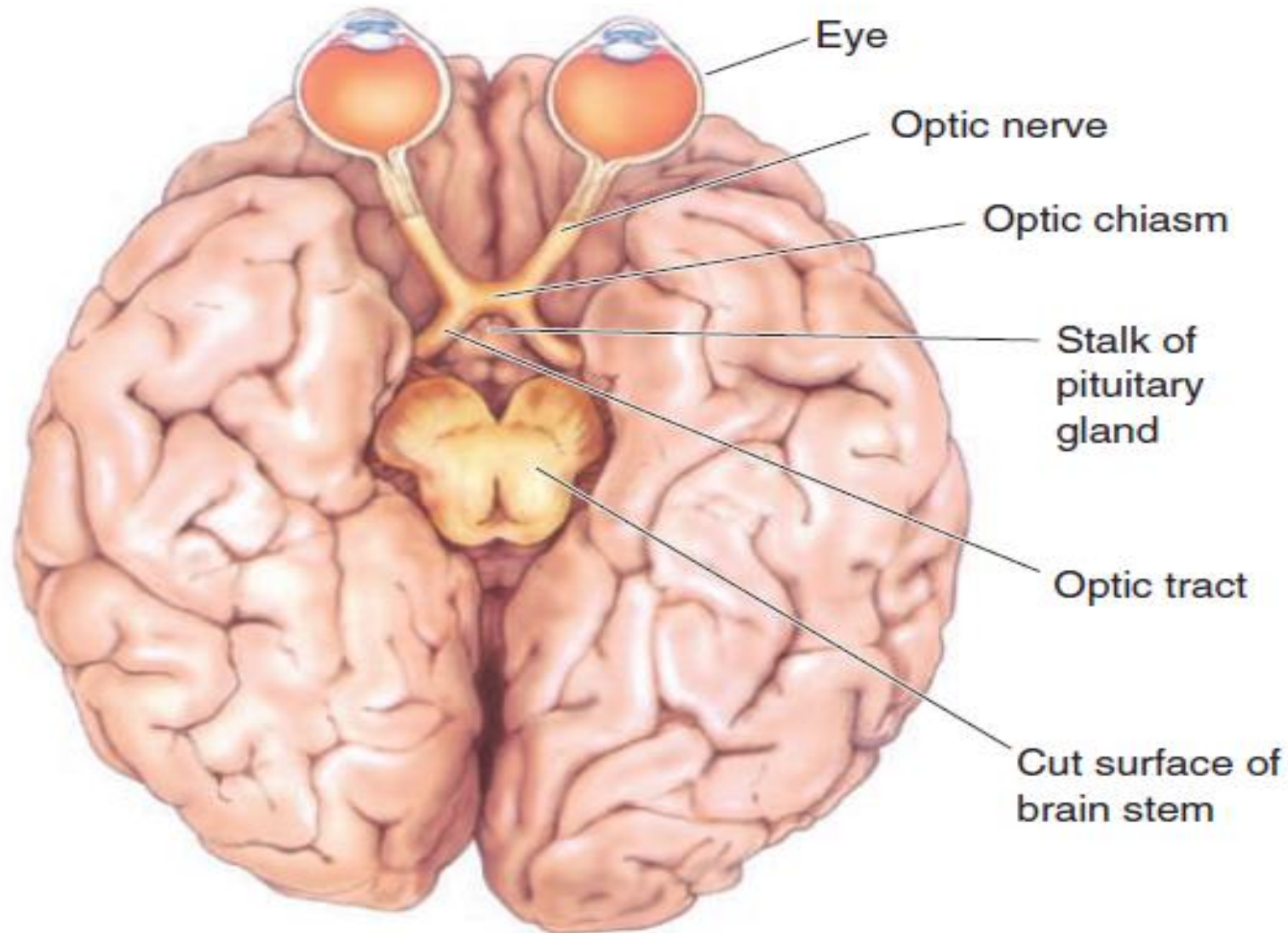
Central Visual System

Visual Pathways

- There are more than ≈ 100 million photoreceptors in the retina
- But only ≈ 1 million axons transmit visual information to the brain
- Ganglion cell axons pass through 3 structures before they form synapses in the brainstem:
 - The optic nerve
 - The optic chiasm and
 - The optic tract
- Retinofugal projection:
 - Neural pathway that leaves the eye, beginning with the optic nerve

Visual Pathways

- Optic nerves:
 - Exit the left and right eyes at the optic disks
 - Travel through the fatty tissue behind the eyes in their bony orbits, then pass through holes in the floor of the skull
- Optic chiasm:
 - Formed by **intersection of optic nerves** from both eyes
 - Lies at the base of the brain (*anterior to the dangling of pituitary gland*)
 - *Optic chiasm is susceptible to pituitary tumors*



Retinofugal Projection

Visual Pathways

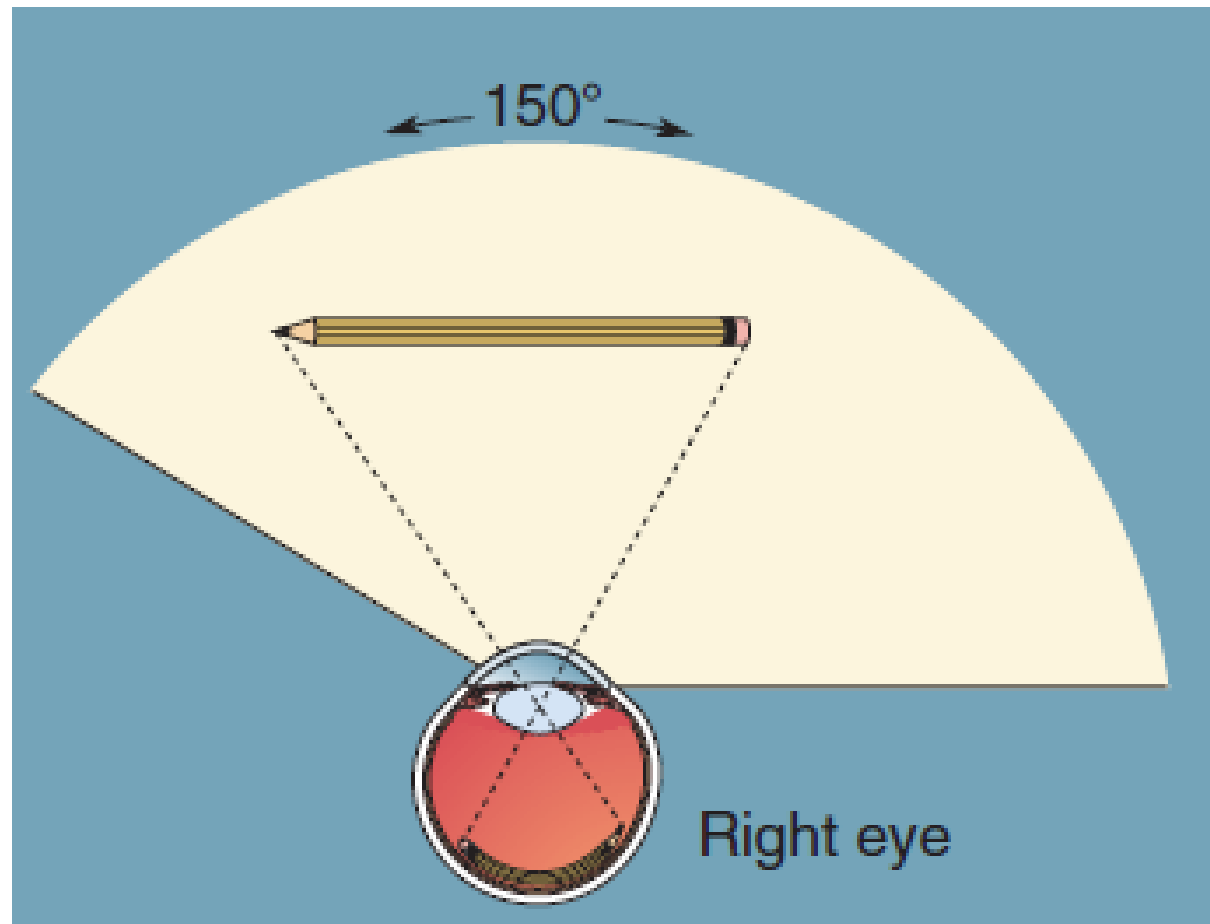
- Optic chiasm:
 - Visual information from each retina is first sorted according to their retinal origin for purposes of **brain lateralization**
 - Visual field information from the left retina will be appreciated in the right hemisphere, and vice versa
 - **Partial decussation** of retinofugal projection occurs at the chiasm
 - Fibers originating in the nasal half of each retina (corresponding to temporal hemifield) cross and enter the contralateral optic tract
 - *Only axons originating in the nasal retinas decussate*

Visual Pathways

- Optic chiasm:
 - Fibers from the temporal half (*corresponding to nasal hemifields*) remain ipsilateral in the optic tract that emerges from the chiasm
 - Thus, each hemisphere receives visual information from both eyes
 - **Fovea** (*point of greatest acuity*) is represented in both hemispheres

Visual Pathways

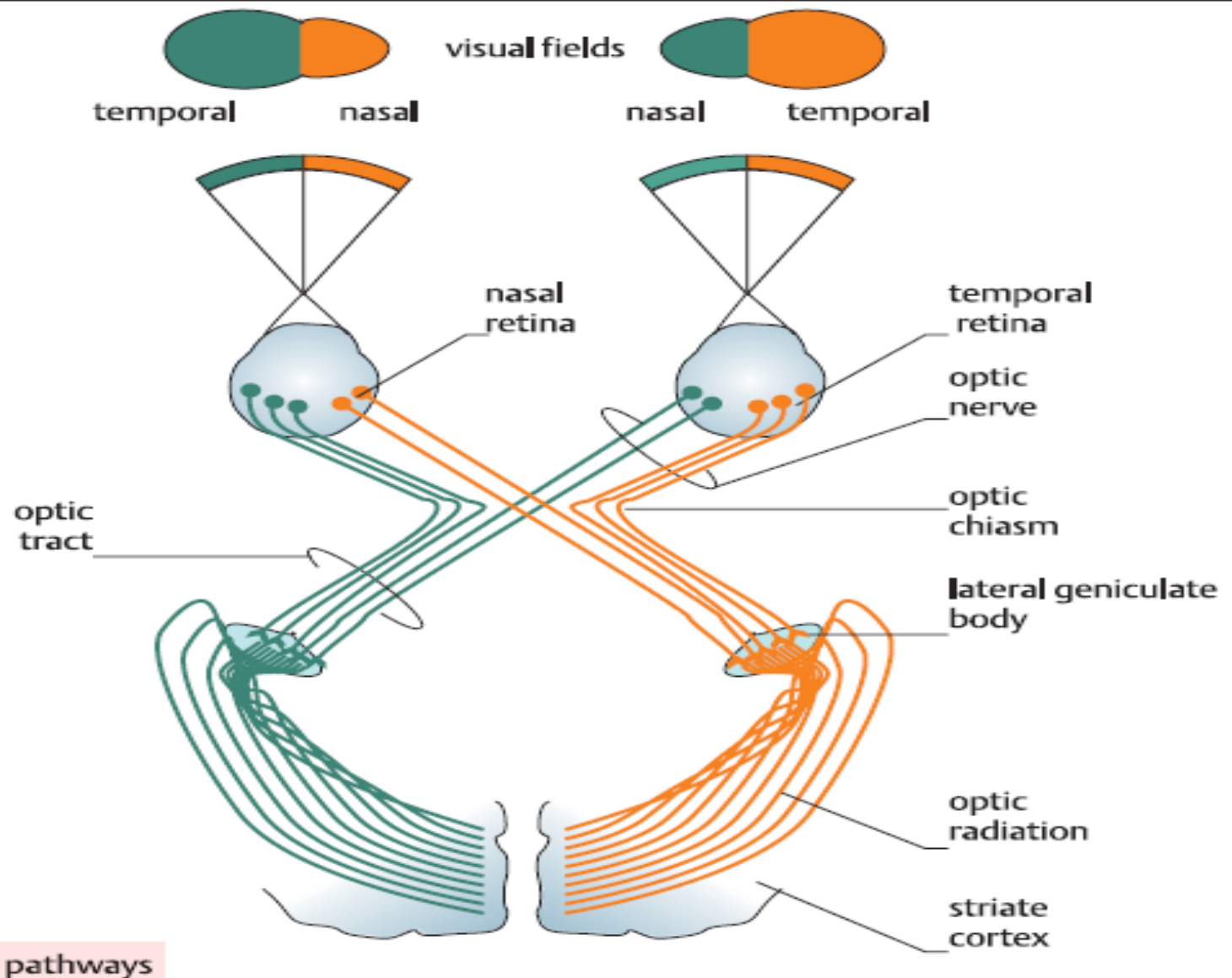
- **Visual field:**
 - The total amount of space that can be viewed by the retina when both eyes are fixated straight ahead
 - The structure and location of the eyes limits how much of the world we can see at any one time
 - **Left visual hemifield** → objects appearing to the left of the midline
 - **Right visual hemifield** → objects appearing to the right of midline
 - **Binocular visual field** → central portion of both visual hemifields which is viewed by both retinas



Visual field

Visual Pathways

- Visual field:
 - Each retina has **nasal** and **temporal fields**
 - Light entering from the left strikes nasal retina of the left eye + temporal retina of the right eye (*and vice versa if from the right*)
 - Thus, objects in the left visual hemifield will be imaged on nasal retina of the left eye and on the temporal retina of the right eye
 - The same principle applies for objects in the right hemifield
 - This pattern of light entry is preserved faithfully throughout the CNS



Visual Pathways

- Visual field:
 - Fibers from the nasal retina crosses at the optic chiasm
 - Fibers from the temporal retina do not cross
 - Thus, all the information about the left visual hemifield is directed to the right side of the brain and vice versa
 - The image of an object in the visual field is **inverted on the retina:**
 - The left visual field is imaged on the right side of the retina
 - The right visual field is imaged on the left side of the retina

Visual Pathways

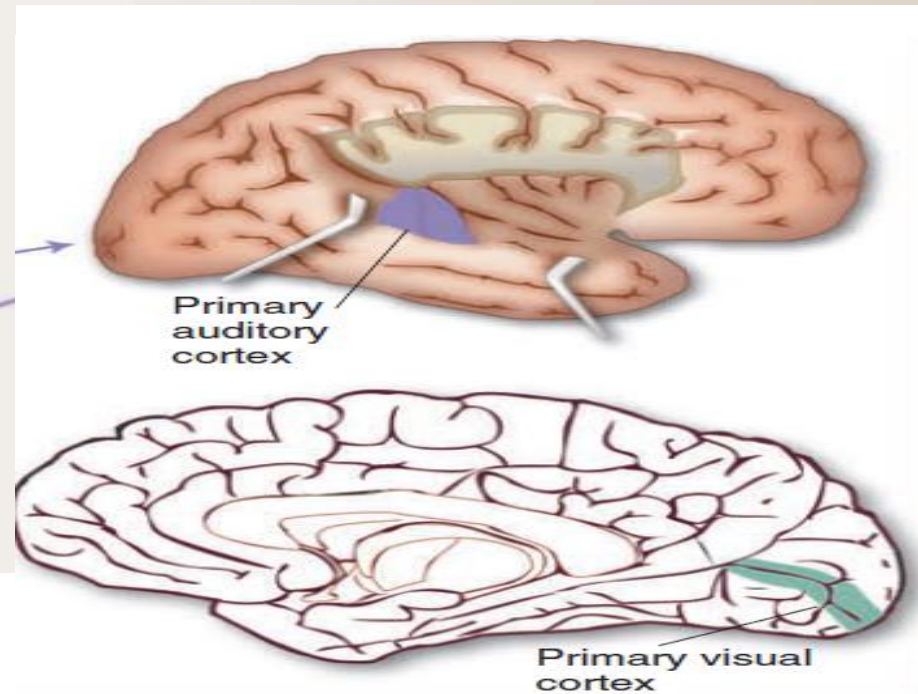
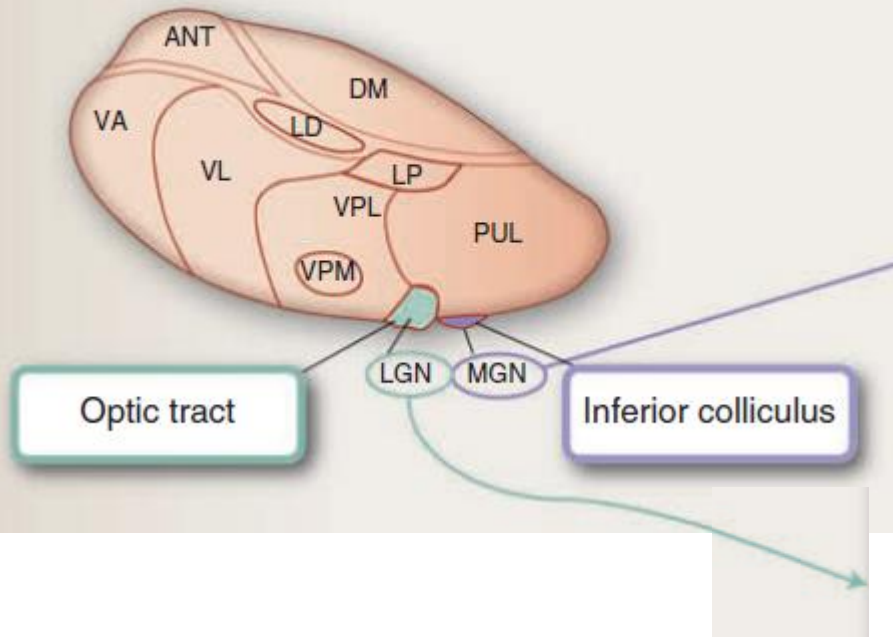
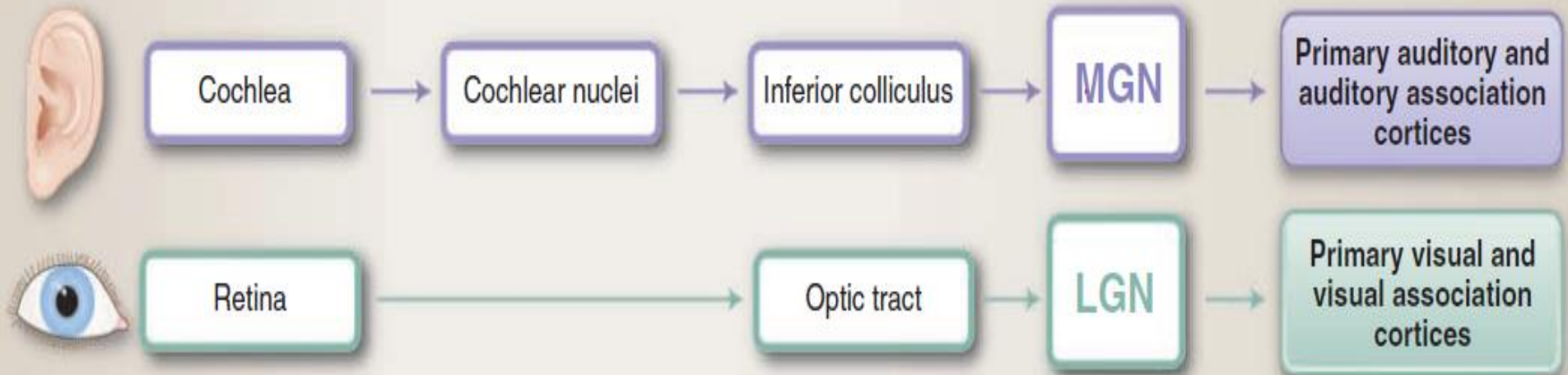
- Optic tracts:
 - Formed from retinofugal axons following partial decussation
 - Run under the pia along the lateral surfaces of the diencephalon
- Targets:
 - Lateral geniculate nucleus (LGN)
 - Suprachiasmatic nucleus (HT)
 - Midbrain (Pretectum + Tectum)
 - Visual cortex...
- These areas are interconnected so that responses are coordinated

Central Visual Systems

- Conscious visual perception involves:
 - **Lateral geniculate nucleus** (LGN) of the thalamus
 - **Primary visual cortex** (area 17, V1, striate cortex)
 - PVC feeds this information to different extrastriate areas in the **temporal & parietal lobes** (*specialized for different types of analysis*)

Central Visual Systems

- Lateral geniculate nucleus:
 - Right and left LGN located in the dorsal thalamus
 - Receive most of the optic fibers
 - Cells have wavelength-dependent receptive fields similar to those of ganglion cells, with concentric on-center and off-center fields
 - Project axons to the primary + association visual cortices
 - Retina → LGN → visual cortex → conscious visual perception
 - Lesions in this pathway causes blindness

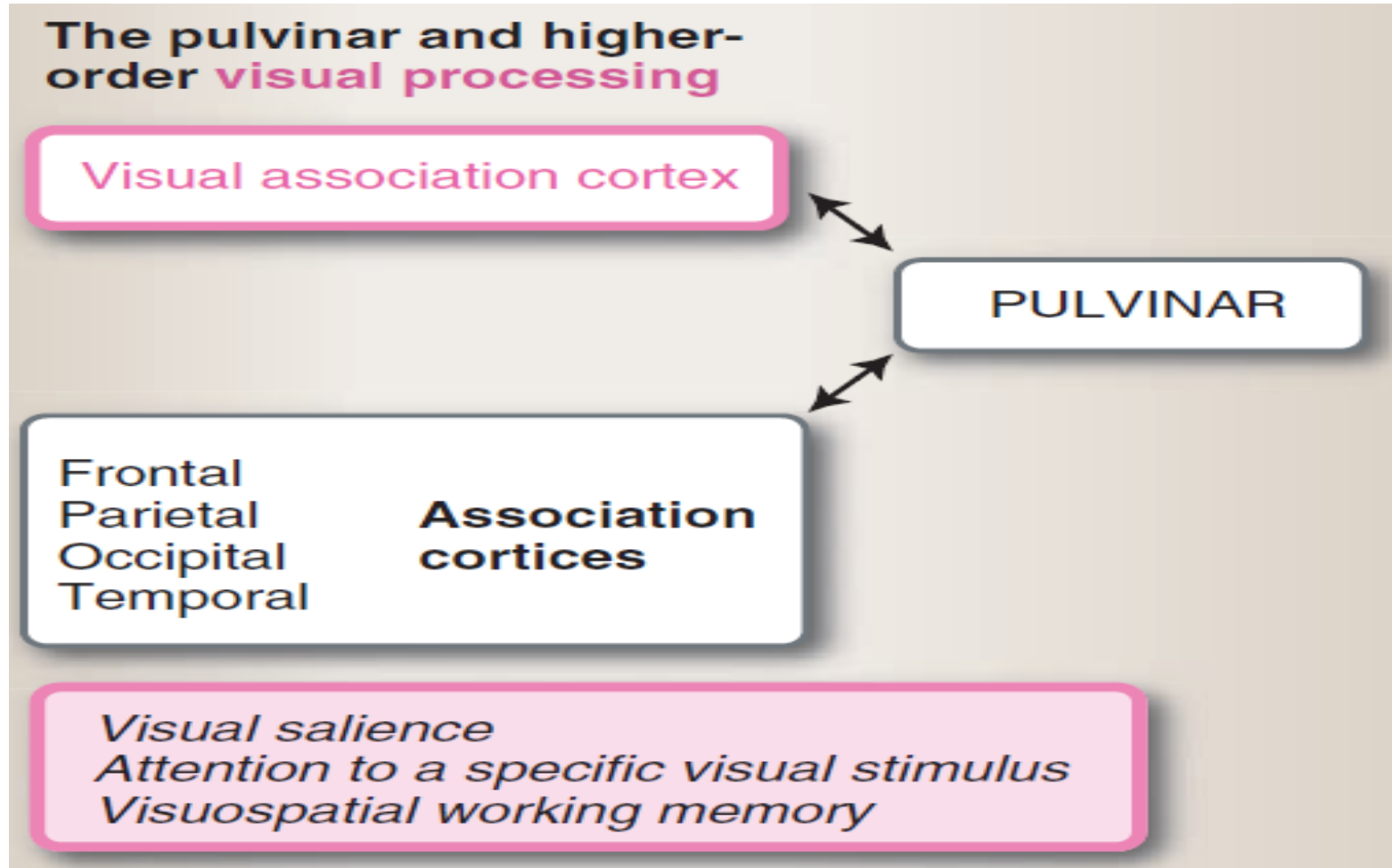


Central Visual Systems

- Superior colliculus:
 - Retinotectal projection
 - Modulates visual reflexes
 - Orienting the eyes in response to new stimuli in the visual periphery
 - Axons from SC projects to the thalamic pulvinar nucleus
 - Pulvinar nucleus also receives input from visual association cortex + cortical association areas...+ → → integration

Central Visual Systems

- Visual processing



Central Visual Systems

- Olivary pretectal and accessory optic nuclei
 - Involved in the pupillary light reflex
 - Control the size of the pupil and certain types of eye movement

Central Visual Systems

- Pupillary light reflex:
 - The pupil contributes to the optical functioning of the eye by continuously adjusting for different ambient light levels
 - Involves connections between the retina and neurons in the brain stem that control the muscles that constrict the pupils
 - This reflex is **consensual** → shining a light into only one eye causes constriction of the pupils of both eyes
 - Constriction of the pupil has the effect of **increasing the depth of focus**

Central Visual Systems

- Pupillary light reflex:
 - When the eye accommodates to the closer point, the image of the farther point on the retina no longer forms a point, but rather a blurred circle
 - Decreasing the aperture (constricting the pupil) reduces the size of this blurred circle so that its image more closely approximates a point
 - In this way, distant objects appear to be less out of focus

Central Visual Systems

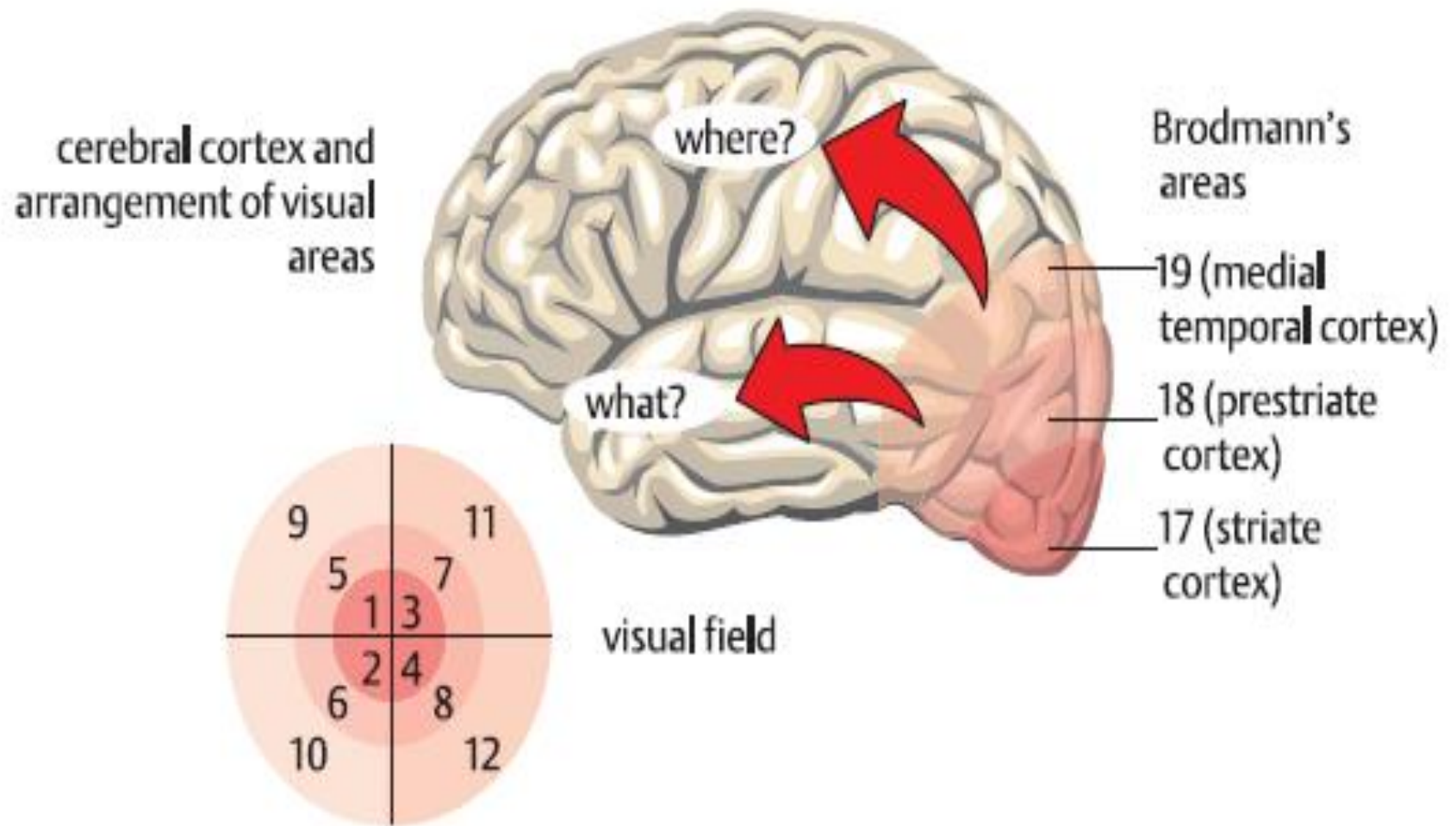
- **Suprachiasmatic nucleus**
 - Play an important role in synchronizing a variety of biological rhythms
 - Governs **diurnal rhythms**
 - Sleep and wakefulness, with the daily dark-light cycle

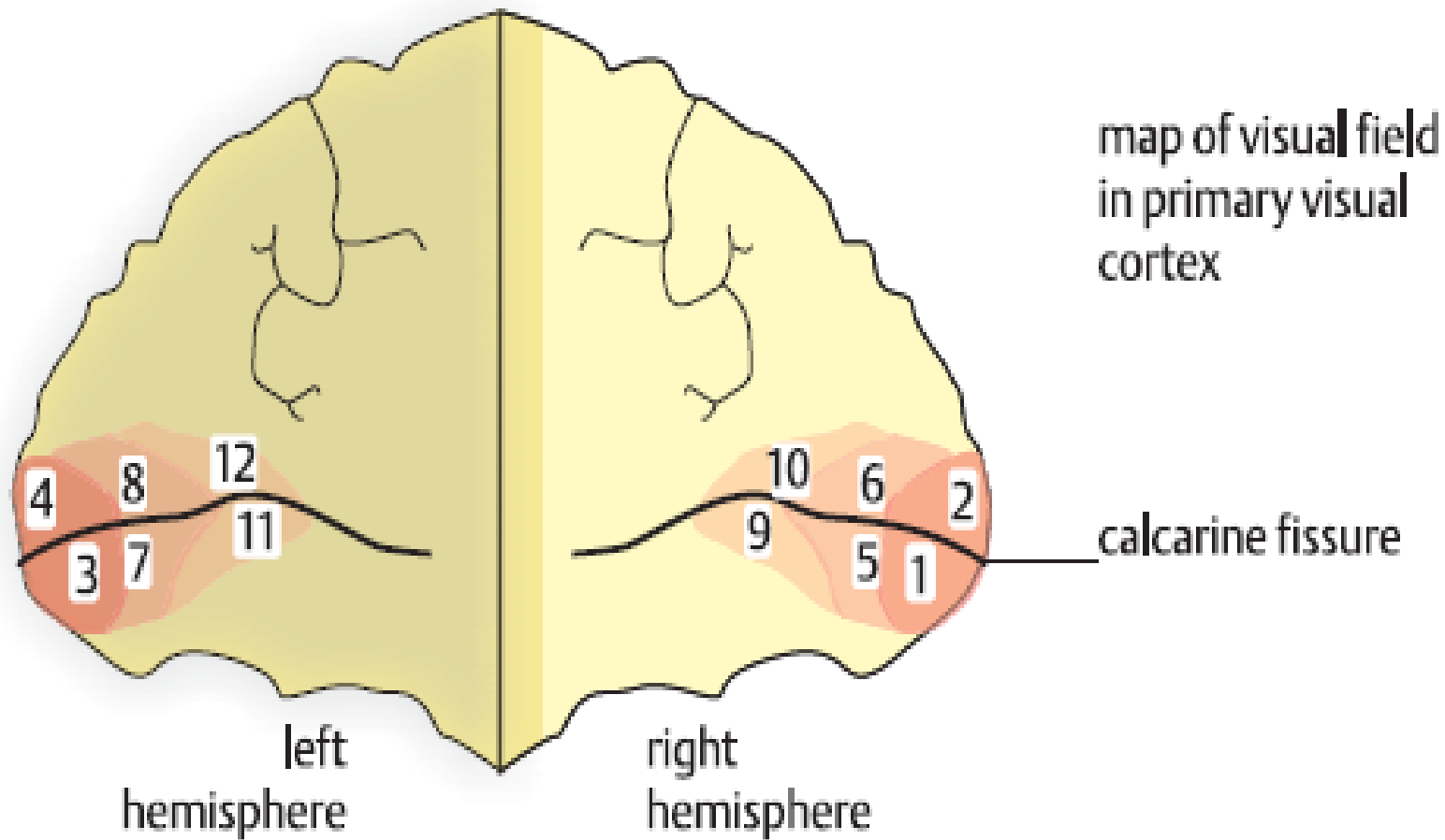
Central Visual Systems

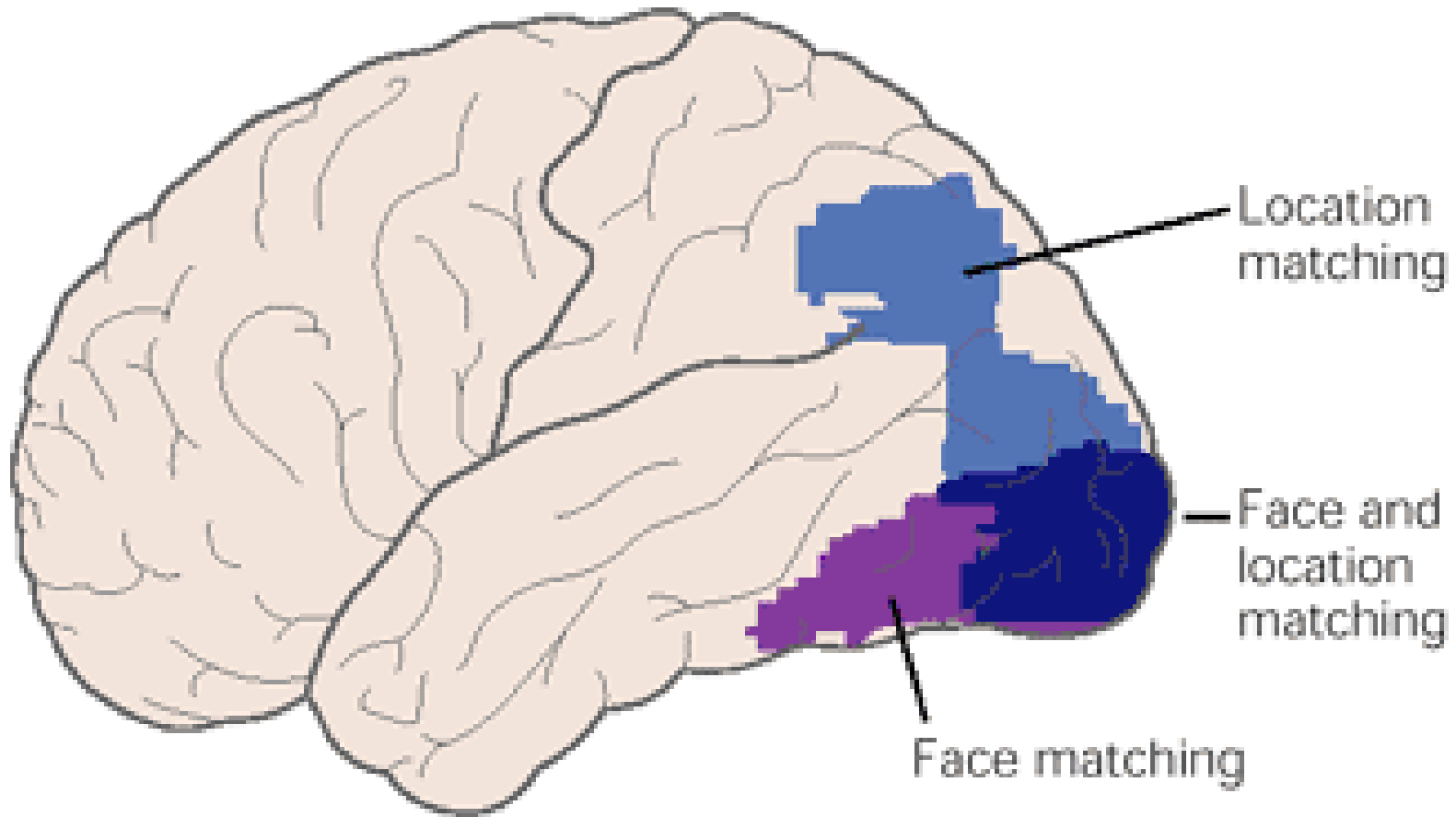
- **Primary Visual Cortex:**
 - Brodmann's area 17/striate cortex
 - Bilateral area lying almost entirely medially on either side of the calcarine sulcus in the occipital lobe
 - Surrounded by visual association areas:
 - Prestriate cortex (Area 18)
 - Medial temporal cortex (Area 19)

Central Visual Systems

- **Primary Visual Cortex:**
 - Receives most of the inputs from the lateral geniculate nucleus
 - Contains an ordered map of the visual field
 - Each hemifield is represented on the contralateral hemisphere
 - On each side, the **upper quadrants** are represented below the calcarine fissure, while the lower quadrants are represented above the calcarine fissure
 - In each hemisphere, a disproportionately large anatomical area of the visual cortex is devoted to the **representation of the fovea**







- Parietal areas → **Localization** of visual inputs
- Temporal areas → **identification** of visual inputs

Summary

- Visual inputs to:
 - Limbic regions (hypothalamus...) subserve motivational behavior related to feeding, sexual activity...
 - Midbrain (pretectum, SC) subserve proprioception & localization
 - Cortex subserve cognitive correlates (recognition...)

Summary

- Visual information:
 - Cortex → detect boundaries and edges of objects for recognition
 - Superior colliculus → locate objects in the external environment
 - Brainstem (pretectum and related nuclei) → assess our position in relation to external objects in the environment
 - Detect movement
 - Assesses the intensity of ambient light (tells the time of day), which involves accommodation and other eye reflexes concerned with the reception of light by the eye

Summary

- **NB:**
 - Transection of **left optic nerve** renders **blind in the left eye only**
 - Transection of the **left optic tract** leads to **blindness in the right visual field** as viewed through either eye
 - **Midline transection** of the optic chiasm leads to **blindness in the peripheral visual fields** on both sides (*regions of the visual field viewed by the nasal retinas*)

